

Education Subsidies as Insurance against Income Risk in Fertility Decisions*

Kanato Nakakuni[†]

December 2025

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Abstract

Empirical studies consistently find that income uncertainty reduces fertility, yet its policy implications remain underexplored. Using a quantitative macroeconomic model, this paper shows that income-tested education subsidies, prevalent across countries, partially insure households against income risk in fertility decisions and generate important aggregate effects. I build and calibrate a lifecycle Aiyagari model with fertility and college enrollment choices, in which altruistic parents make asset transfers to support their children's college education. Having more children increases the risk that parents may be unable to finance their children's education after bad income shocks or must endure substantially lower lifetime consumption to do so. This mechanism discourages fertility under income uncertainty, generating the observed negative relationship between income risk and fertility. Counterfactual experiments show that income-tested college subsidies insure households against this risk associated with having children. The resulting fertility responses amplify the macroeconomic effects of the education subsidy—including reductions in income inequality, welfare gains, and increases in education and income—through intergenerational linkages and general-equilibrium feedback, both in the long run and along the transition path.

Keywords: Fertility, incomplete market, college subsidy, income risk

JEL codes: C68, I28, J13, J24

* I am grateful to Minsu Chang, Matthias Doepke, Paula Gobbi, James Graham, Jeremy Greenwood, Yusuke Ishihata, Akihisa Kato, Sagiri Kitao, Tomoaki Kotera, Dirk Krueger, Minjoon Lee, Alexander Ludwig, Kazushige Matsuda, Toshihiko Mukoyama, Makoto Nakajima, Tomoyuki Nakajima, Makoto Nirei, Lukas Nord, Yena Park, Ponpoje Porapakkarm, Xavier Ragot, Víctor Ríos-Rull, Cezar Santos, Kamila Sommer, Yuta Takahashi, Satoshi Tanaka, Kazuhiro Teramoto, Michèle Tertilt, Hanna Wang, Tomoaki Yamada, Shintaro Yamaguchi, Fang Yang, Minchul Yum, Anson Zhou, and participants at numerous seminars and conferences for their many helpful comments. This paper was previously circulated as “Education Subsidy as Insurance for Fertility Choices.” This study is financially supported by the Japan Society for the Promotion of Science (23KJ0374). I also acknowledge support from the German Research Foundation (CRC-TR-224 project A3 and Gottfried Wilhelm Leibniz-Prize, TE 966/2-1).

[†]University of Mannheim, Email: nakakunik@gmail.com

1 Introduction

Total fertility rates have reached historic lows, contributing to population decline and a shrinking share of the working-age population. Low fertility therefore raises macroeconomic concerns such as labor force contraction, the sustainability of social security systems, and lower living standards. Among the determinants of fertility, labor market factors—such as job flexibility and income uncertainty—have become increasingly important in explaining the cross-country variation in fertility (Doepke et al., 2023). In particular, empirical studies consistently find that income uncertainty reduces fertility (e.g., Chabé-Ferret and Gobbi, 2018). Despite the consensus on the importance of income uncertainty in shaping fertility choices, little is known about its policy implications and associated macroeconomic consequences.

This paper addresses this gap and makes three main contributions. First, I analytically and quantitatively show that income-tested college subsidies, prevalent across countries, act as partial insurance against income risk in fertility choices and thereby increase fertility. Second, I show that the resulting increases in fertility amplify the macroeconomic effects of college subsidies—such as reductions in inequality, welfare improvements, and higher income levels—through general-equilibrium (GE) effects and intergenerational linkages. Third, this study is the first to construct a GE lifecycle model that jointly integrates fertility and college enrollment choices.

I start with a simple analytical model of fertility choices with uninsurable income risk to highlight an insurance effect of income-tested subsidies on fertility. In this model, ex-ante identical households make fertility choices under income uncertainty, with spending on children assumed to be fixed and exogenously given, and income levels are realized after fertility choices are made. Having children can ex-post become very costly in utility terms following a bad income realization because, if they have more children, a bad income shock results in significantly lower consumption (or, lower spending on children if it were endogenous) compared to having fewer children, leading to higher marginal utilities. However, parents cannot “reduce” the number of children after facing a bad income shock because fertility choices are irreversible. Therefore, income uncertainty may deter parents from having another child, lowering fertility rates.

I first show that a mean-preserving spread of the income distribution (i.e., a rise in income uncertainty) in the model leads to lower fertility, providing a theoretical foundation for the empirical finding that higher income uncertainty lowers fertility.¹ I then consider an income-tested subsidy that redistributes income from households with positive (good) income shocks to those with negative (bad) shocks, ensuring that the expected value of the subsidy received is zero. This design rules out the subsidy’s effects on fertility driven by lower expected monetary costs of children (or the lower “price” of children), which I refer to as the *price effect*. I show that despite the absence of this effect, income-tested subsidies still increase fertility: they smooth marginal utilities conditional on having chil-

¹See Modena et al. (2014); Chabé-Ferret and Gobbi (2018); De Paola et al. (2021); Guner et al. (2024). For example, Chabé-Ferret and Gobbi (2018) identify the causal effect of increased economic uncertainty—measured by the volatility of income growth rates—on cohort fertility by exploiting within state-cohort cell variation in economic conditions using twentieth-century U.S. data. Although parents may choose to postpone childbearing to a later stage in life, this ultimately results in lower (completed) fertility rates, as the ability to bear children declines with age (Sommer, 2016).

dren, thereby lowering the expected marginal costs of children, even though the expected monetary costs remain unchanged. I refer to this fertility increase as the *insurance effect* of the subsidy.

To evaluate the quantitative importance of this insurance effect relative to other potential effects, which is informative for policy design, and to examine the macroeconomic effects associated with these fertility margins, I develop a lifecycle Aiyagari model (i.e., an incomplete market, GE lifecycle model) with several key elements: (1) fertility choices about how many children to have, (2) college enrollment choices, (3) intergenerational linkages between parents and children through asset transfers and exogenous transmission of innate abilities and tastes for education, and (4) imperfect substitutability among workers with different educational backgrounds in the aggregate production function.

The model is calibrated using data from the Japanese Panel Survey of Consumers (JPSC). To track completed fertility and children’s education outcomes, I use the oldest cohort in the data, which was born between 1959 and 1969. Japan is a unique environment for the following reasons, making it a useful case study. Most importantly, the Japanese government has provided less financial support for higher education compared with other countries (Fig A.1). The annual average expenditure—mainly tuition fees—by a college student is approximately 2 million yen, which corresponds to almost half of the median household annual income.² Of this amount, parental transfers cover around 60% on average, while the remainder is financed through student labor supply or government-backed student loans.

The expensiveness of college education strongly affects both college enrollment and fertility decisions. Among individuals who did not attend college and whose parental income fell within the first quintile, 60% answered that “it was difficult to proceed to college due to financial reasons,” whereas the corresponding shares were approximately 35% and below 10% for the third and fifth quintiles, respectively (Hamanaka, 2017). In addition, descriptive evidence suggests that the cost of college education—borne primarily by parents rather than by students or the government—creates a significant barrier to having more children (Appendix A). Given that the college enrollment rate in Japan has increased over time and recently exceeded 50%, and that more than 80% of parents desire their children to proceed to college (Fig A.5), expensive college education is a relevant problem for most parents. In response, the Japanese government introduced an income-tested college subsidy in 2020 as part of its pro-natal policy package and has discussed a gradual expansion of its coverage.³ Therefore, Japan provides a useful setting to examine the role of education subsidies not only in college enrollment decisions but also in fertility choices and the interaction between these two margins.

In the model, wages are specified as a function of age, education, innate ability, and a stochastic component. Using the data, I estimate a first-order autoregressive process for the stochastic component, characterized by two parameters, separately by education group. Taking these estimates and other parameters obtained directly from the data as given, I then jointly calibrate the remaining model parameters to minimize the distance between empirical and model-implied moments.

The calibrated model replicates key cross-sectional moments across the income distri-

²The median household income in 2021 was 4.4 million yen, according to the Comprehensive Survey of Living Conditions by the Ministry of Health, Labour and Welfare (MHLW).

³See Appendix A for details.

bution, including fertility, spending on children, parental transfers for college, and enrollment rates, as well as intergenerational persistence in income and education. Importantly, the model also replicates several measures of fertility elasticities as non-targeted moments: (1) the fertility elasticity with respect to government cash transfers to households with children, as reported in previous empirical studies (e.g., [Milligan, 2005](#); [Malkova, 2018](#)); (2) the *fertility-subsidy gradient*, capturing the relationship between fertility rates and the level of education subsidies, obtained using cross-country panel data; and (3) fertility differentials between college- and high school-graduate parents.

In the benchmark model, college students can access government-provided student loans that mimic the actual loans available in Japan, which are income- and ability-tested, while grants are unavailable, reflecting the situation in Japan before 2020.⁴ I then conduct several experiments using the model, including the introduction of income-tested grants implemented in 2020. These grants provide cash transfers to college students from households in the bottom 15% of the income distribution, and per-student grants amount to two-thirds of the average expenses of college students.

The main findings are summarized as follows. First, the direct effects of introducing the income-tested subsidy—defined as the effects of the subsidy while holding prices, tax rates, and household distribution constant—indicate a 2.1% increase in fertility. A decomposition analysis reveals that insurance effects play a more significant role than price effects: 62% of the direct effects on fertility are attributed to the insurance effect, while 38% are due to the price effect.

Next, I also find that these fertility responses amplify the policy’s effects on inequality reduction, welfare improvement, and raising education and output levels in the long run through the following mechanism. First, the positive effect on fertility is more significant among college graduates since their children are more likely to attend college and are more likely to be eligible for the subsidy. Given intergenerational linkages via abilities, tastes, and assets, a higher fertility among college graduates implies that the children born due to the policy are more likely to attend college. Therefore, the policy further increases college enrollment rates through this fertility channel in the long run. A higher college enrollment rate reduces the skill premium and contributes to greater accumulation of aggregate human capital, thereby expanding the tax base and mitigating the increase in tax rates needed to finance the policy. These effects (i.e., a higher probability of attending college, higher expected income, lower inequality, and lower taxes), induced by the fertility responses and associated GE effects, generate more significant welfare gains for newborn agents under the veil of ignorance. This amplification is highlighted by comparing the results with those under an *exogenous fertility* setup, both at the steady state and during transitional dynamics.

To summarize, I show that (1) income-tested subsidies provide insurance for fertility choices under income uncertainty and (2) the resulting fertility responses amplify the macroeconomic effects of the policy. Although the model is applied to Japanese data, the underlying mechanisms, highlighted through both analytical and quantitative models, are not unique to Japan but offer insights applicable to other countries as well.

⁴This model does not allow students to access financial aid (grants or loans) provided by institutions besides the government (e.g., colleges or other private entities). In 2018, among students in four-year universities who used any financial aid, 83.8% used only government-provided aid, 8.6% used only non-government-provided aid, and 7.6% used both, according to the Student Life Survey (SLS, 2018).

2 Related Literature

This paper contributes to several strands of literature. The first is the macroeconomic analysis of college education subsidies.⁵ The literature highlights important implications of the college subsidies, including the GE feedback through an increased supply of college-educated workers (e.g., [Krueger and Ludwig, 2016](#)), the crowding-out of parental transfers and student labor supply that may dampen policy effects ([Abbott, Gallipoli, Meghir, and Violante, 2019](#)), and the complementarity between college subsidies and policies promoting pre-college human capital accumulation that affects college completion success ([Krueger, Ludwig, and Popova, 2024](#)). I contribute to this literature by incorporating fertility choices into an otherwise standard framework—an incomplete-markets, GE lifecycle model with college enrollment and intergenerational linkages—and by highlighting an additional channel through which college subsidies operate: their insurance role in fertility decisions and the resulting amplification of policy effects along the fertility margin.

Second, this paper builds on macroeconomic studies based on the quantity-quality trade-off framework.⁶ In particular, this study is closely related to three papers in the literature that consider the macroeconomic effects of education policies using models with the quantity and quality of children: [De la Croix and Doepke \(2004\)](#), [Zhou \(2022\)](#), and [Kim, Tertilt, and Yum \(2023\)](#). These studies highlight the important roles of (1) fertility differentials in understanding the macroeconomic effects of private and public schooling regimes ([De la Croix and Doepke, 2004](#)), (2) the effectiveness of public education expenditures in boosting fertility compared to other pro-natal policies, along with their broader macroeconomic implications ([Zhou, 2022](#)), and (3) the externalities of parental investments in evaluating pro-natal policies and education taxes, as well as in understanding the underlying causes of low fertility in Korea ([Kim, Tertilt, and Yum, 2023](#)). The contribution of this study, and what differentiates it from the existing literature, is that it demonstrates the insurance role of education subsidies in fertility choices. From a modeling perspective, the closest work is [Daruich and Kozlowski \(2020\)](#). They develop a partial equilibrium lifecycle model with college enrollment, fertility, and inter-vivos transfers to investigate the role of fertility choices and family transfers in explaining intergenerational mobility in the U.S. Building on their work, this study extends their framework by adopting a GE approach to examine the macroeconomic implications of education policies.

Third, this study is also related to the literature on fertility choices in incomplete market models and, in general, on the implications of heterogeneous family structures under incomplete markets.⁷ Previous studies demonstrate that having a child can be considered making “consumption commitments,” and that income volatility thus makes households hesitate to have children, leading to a lower fertility ([Sommer, 2016](#); [Guner, Kaya, and Sánchez-Marcos, 2024](#)).⁸ This study contributes to the literature by examining

⁵For example, [Krueger and Ludwig \(2016\)](#); [Abbott, Gallipoli, Meghir, and Violante \(2019\)](#); [Matsuda and Mazur \(2022\)](#); [Krueger, Ludwig, and Popova \(2024\)](#).

⁶For example, [De La Croix and Doepke \(2003\)](#); [De la Croix and Doepke \(2004\)](#); [Manuelli and Seshadri \(2009\)](#); [Greenwood, Guner, Kocharkov, and Santos \(2016\)](#); [Cordoba, Liu, and Ripoll \(2019\)](#); [Daruich and Kozlowski \(2020\)](#); [Zhou \(2022\)](#); [Kim, Tertilt, and Yum \(2023\)](#); [Kitao and Nakakuni \(2024\)](#).

⁷For example, [Cubeddu and Ríos-Rull \(2003\)](#); [Schoonbroodt and Tertilt \(2014\)](#); [Vandenbroucke \(2014\)](#); [Santos and Weiss \(2016\)](#); [Sommer \(2016\)](#); [Bacher, Grübener, and Nord \(2024\)](#); [Coskun and Dalgic \(2024\)](#); [Guner, Kaya, and Sánchez-Marcos \(2024\)](#); [Adamopoulou, Hannusch, Kopecky, and Obermeier \(2025\)](#).

⁸This effect is particularly pronounced in the early stages of life, leading to delayed fertility. Since the

policies that can potentially insure against the risks associated with having a child and highlights the insurance effects of income-tested college subsidies on fertility choices.

Finally, this study contributes to the literature on pro-natal policies using structural models⁹ by adding information about the effects of college subsidies, a novel pro-natal policy implemented in Japan. Compared with typical pro-natal transfers such as baby bonuses, the notable feature of education subsidies is that it increases the average human capital by promoting skill acquisition.¹⁰ By examining this novel policy in Japan, this study provides insights into countries considering countermeasures against macroeconomic challenges of low fertility.

3 Analytical Model

This section constructs a simple analytical model of fertility choices with uninsurable income risk to highlight the insurance effect of income-tested subsidies on fertility.

Setup: Consider a decision problem where ex-ante identical households choose the number of children $n \geq 0$ and consumption $c > 0$, facing uninsurable income risk, to maximize their expected utility, $\mathbb{E}U(n, c) = b(n) + \mathbb{E}V(c)$. For simplicity, this model assumes that the number of children is a continuous variable and needs to be positive. The household first chooses the number of children n , before an income level I is realized. After making the fertility choices, an income level is realized, which is drawn from a distribution $F(I)$. The price of the consumption goods is normalized to one, and having children takes a fixed cost $\underline{a}$, net of the government subsidy $\bar{g}(I)$. Let $a(I) = \underline{a} - \bar{g}(I)$ be the (net) unit costs of having a child. The household problem is formulated as follows:

$$\begin{aligned} \max_{n > 0} \quad & \{b(n) + \mathbb{E}_I[V(c)]\} \\ \text{s.t.} \quad & c + n a(I) = I, \\ & a(I) = \underline{a} - \bar{g}(I), \\ & I \sim F(I). \end{aligned}$$

The utility functions for consumption V and for number of children b satisfy the standard properties as follows: (1) $V' > 0$, (2) $V'' < 0$, (3) $V''' > 0$, (4) $b' > 0$, and (5) $b'' < 0$.

ability to reproduce declines with age, delayed fertility ultimately results in lower fertility rates (Sommer, 2016). Similarly, Santos and Weiss (2016) show that an increase in income volatility leads to delays in marriage, as marriage often entails a consumption commitment in the form of raising children. Coskun and Dalgic (2024) develop an overlapping generations model incorporating the quantity-quality trade-off and highlight the importance of accounting for gender differences in income volatility, as well as in income levels, within households in understanding the cyclical behavior of fertility.

⁹Previous studies examine the effects of childcare subsidization (e.g., Bick, 2016), cash transfers (e.g., Kim, Tertilt, and Yum, 2023; Nakakuni, 2024), both of them (e.g., Zhou, 2022; Hagiwara, 2025), parental leave policies (e.g., Erosa, Fuster, and Restuccia, 2010; Yamaguchi, 2019; Wang, 2022; Kim and Yum, 2023), and tax reform (e.g., Jakobsen, Jørgensen, and Low, 2022).

¹⁰Previous studies show that pro-natal transfers would lower aggregate human capital because they make parents shift from the “quality” toward “quantity” of children (e.g., Zhou, 2022; Kim et al., 2023). Indeed, Appendix E.(iv) shows that the income-tested grants would lead to greater aggregate human capital and output than pro-natal transfers conducted in an expenditure-neutral way.

Effects of Income Risk on Fertility: I first show that an increase in income risk—in terms of a mean-preserving spread of income distribution $F(I)$ —lowers the fertility n .¹¹ To this end, I assume that income can be either high or low, that each occurs with probability 50% (i.e., income is uniformly distributed), and that the income realization is therefore represented as follows:

$$I = \begin{cases} \bar{I} + \nu & \text{w.p. } 1/2 \\ \bar{I} - \nu & \text{w.p. } 1/2 \end{cases}$$

Here, $\nu < \bar{I}$ is a parameter that governs the dispersion of income, and $\bar{I} > 0$ is the mean of the income distribution. Also, because the focus is now on the pure effect of income uncertainty on fertility, I do not consider the income-tested subsidy; the cost of having children is $\underline{a}$ for everyone. Given this setup, the following proposition follows.

Proposition 1. *Let n be the optimal number of children. It holds that*

$$\frac{dn}{d\nu} < 0$$

Namely, a mean-preserving spread of the income distribution (i.e., an increase in income risk) reduces fertility.

Proof. I first take the first-order condition with respect to n as follows:

$$\underline{a}^{-1} \cdot b'(n) = \frac{1}{2} [V'(\bar{I} + \nu - n \cdot \underline{a}) + V'(\bar{I} - \nu - n \cdot \underline{a})] \quad (1)$$

I then express n as a function of ν and take the first-order derivatives of the both sides of equation (9) with respect to ν . After rearranging the terms and solving for $dn/d\nu$, I obtain the stated result:

$$\frac{dn}{d\nu} = \frac{\overbrace{V''(\bar{I} + \nu - \underline{a} \cdot n(\nu)) - V''(\bar{I} - \nu - \underline{a} \cdot n(\nu))}^{>0 \text{ (since } V''' > 0)}}{\underbrace{2\underline{a}^{-1}b''(n(\nu)) + \underline{a} \cdot [V''(\bar{I} + \nu - \underline{a} \cdot n(\nu)) + V''(\bar{I} - \nu - \underline{a} \cdot n(\nu))]}_{<0 \text{ (since } b'' < 0 \text{ and } V'' < 0)}} < 0$$

□

The intuition and mechanism are analogous to the reduction in consumption caused by precautionary saving. The key assumption underlying this proposition is that the third derivative of utility with respect to consumption is positive, which is the central condition that generates precautionary saving.

When households choose the number of children, they do not yet know which income level will be realized once the children are born. If they increase fertility marginally, the marginal utility of consumption becomes particularly high in states where income turns out to be low. This is because childbearing is an irreversible choice—they cannot reduce the number of children after observing a bad income realization—and they must

¹¹The negative effect of income uncertainty on fertility is supported by empirical studies ([Chabé-Ferret and Gobbi, 2018](#); [Guner et al., 2024](#)).

allocate resources to their children even when income is low. Consequently, households are cautious about having too many children when making fertility decisions under income uncertainty. This precaution leads them to limit fertility, thereby reducing the number of children they choose to have.

Effects of Income-tested Subsidies on Fertility under Income Uncertainty:

Now I study the effects of income-tested subsidies on fertility under income uncertainty. I consider the general income distribution $F(I)$ and a subsidy scheme $\bar{g}(I)$. The following assumptions on functional forms are imposed.

First, the utility function of children $b(n)$ takes the form of the constant relative risk aversion utility function, $n^{1-\sigma}/(1-\sigma)$, with some positive value $\sigma > 0$. Second, the subsidy function $\bar{g}(I)$ takes the form

$$\bar{g}(I; \varepsilon) = (\mathbb{E}I - I) \cdot \varepsilon,$$

where $\varepsilon \geq 0$ is a policy parameter that governs the degree of the redistribution. Note that this subsidy function $\bar{g}(I; \varepsilon)$ implies that (1) the subsidy is income-contingent (or income-tested), where households with below-average income realizations receive a positive subsidy, while those with above-average income realizations receive a negative subsidy or incur a “tax,” and (2) the expected cost of children remain constant at $\underline{a}$ because the expected value of the subsidy is zero for any ε (i.e., $\mathbb{E}\bar{g}(I; \varepsilon) = \mathbb{E}(\mathbb{E}I - I) \cdot \varepsilon = 0$). For convenience, I express the optimal choices for consumption and fertility as functions of the parameter ε , $c(\varepsilon)$ and $n(\varepsilon)$. I define the parameter space for ε as $\mathcal{E} \equiv \{\varepsilon \mid 1 - \varepsilon \cdot n(\varepsilon) > 0 \text{ \& } \varepsilon \geq 0\}$, which ensures that higher (lower) income realizations correspond to higher (lower) consumption (i.e., $\partial c / \partial I > 0$ and thus $\partial V'(c) / \partial I < 0$).

The first-order condition implies that the optimal fertility choice is characterized as:

$$n(\varepsilon) = \{\mathbb{E}[a(I; \varepsilon) \cdot V'(c(\varepsilon))]\}^{-\frac{1}{\sigma}} \quad (2)$$

This means that the fertility rate decreases with the expected marginal costs of having children, which comprise both the monetary costs of children and the utility costs associated with foregone consumption. Note that the first-order derivative of the optimal fertility choice n with respect to ε measures how fertility responds to an increase in the extent of redistribution embedded in income-tested subsidies. In particular, evaluating this derivative at $\varepsilon = 0$ captures the fertility response to the introduction of an income-tested subsidy, since $\varepsilon = 0$ represents a scenario with no subsidy. I obtain the following result regarding the effect of income-tested subsidies on fertility.

Proposition 2. *Let $n(\varepsilon)$ be the optimal number of children given the redistribution policy parameter $\varepsilon \in \mathcal{E}$. It holds that:*

$$n'(0) \equiv \left. \frac{dn(\varepsilon)}{d\varepsilon} \right|_{\varepsilon=0} > 0,$$

which means that the introduction of an income-tested subsidy increases fertility.

Proof. See Appendix B.(ii).(B). □

Here, I explain the intuition behind why this income-tested subsidy increases fertility, drawing primarily on the optimality condition given in equation (2). Note that equation (2) can be rewritten as

$$n(\varepsilon) = \left(\frac{1}{\underbrace{\underline{a} \cdot \mathbb{E}V'(c(\varepsilon))}_{(i)} - \underbrace{\text{cov}(g(I; \varepsilon), V'(c(\varepsilon)))}_{(ii)}} \right)^{1/\sigma} \quad (3)$$

Having an additional child is costly in utility terms because it requires households to forgo consumption that they could otherwise enjoy. This marginal cost of having children consists of the expected monetary costs and the expected marginal utility from consumption, which is captured by the term (i) in equation (3). Recall that the expected monetary costs are identical across different subsidy schemes characterized by different $\varepsilon \in \mathcal{E}$ since $\mathbb{E}\bar{g}(I; \varepsilon) = \mathbb{E}(\mathbb{E}I - I) \cdot \varepsilon = 0$. Therefore, it is only the expected marginal utility from consumption ($\mathbb{E}V'(c(\varepsilon))$) that makes a difference across different subsidy schemes in this setup.

If they have more children, a bad income shock is more costly in marginal utility because, holding the monetary costs of children constant, it leads to lower consumption. Given that having children is an irreversible choice, households cannot adjust the number of children after observing an income level. If a subsidy $\bar{g}(I; \varepsilon)$ is available, such costs of having children in marginal utility from consumption become lower because they can receive some cash transfers after facing a bad income shock, which increases fertility. This effect is captured by the covariance term (ii) in equation (2); the marginal utility of consumption and the subsidy co-move positively, because a higher (lower) income realization reduces (increases) both the marginal utility and the subsidy received, given $\varepsilon \in \mathcal{E}$. Note that this effect, represented by the covariance term (ii), disappears if there is no subsidy (i.e., $\varepsilon = 0$), because $\bar{g}(I; 0)$ is constant (zero) for any income I .

To recap, the intuition behind the higher fertility due to the income-tested subsidies is that (1) having children can be significantly costly ex-post in terms of foregone consumption and the resulting higher marginal utility, and (2) an income-tested subsidy smooths the marginal utility to some extent, which we can interpret it as partial insurance against the high (marginal) utility costs of children induced by income shocks.

Next steps: This section presents a simple analytical model and demonstrates that income-tested subsidies can positively impact fertility by providing insurance for fertility choices under income uncertainty. This raises further questions to explore: (1) to what extent do real-world income-tested education subsidies provide such insurance? and (2) what are the implications of the resulting fertility responses for other macroeconomic objectives, such as aggregate output and income distribution? The current model is too simple to answer these questions in that it abstracts several critical aspects, such as (i) more realistic income process over the lifecycle, (ii) saving and labor supply decisions as self-insurance against income risk, (iii) endogenous choices on how much money to spend on children, and (iv) market production and the general equilibrium setup, which allows us to examine the macroeconomic implications. The next section thus constructs a quantitative macroeconomic model that incorporates those ingredients.

4 Quantitative Model

I incorporate fertility choices into an otherwise standard framework in the macroeconomic literature of college subsidies: an incomplete market, general equilibrium (GE) lifecycle model with college enrollment choices and intergenerational linkages. Section 4.1 provides an overview of the lifecycle of this economy. Section 4.2 elaborates on the primitives of the model, and then Section 4.3 formulates the household’s problems. Section 4.4 discusses the equilibrium of this economy.

4.1 Overview of the lifecycle

Figure 1 illustrates the key stages—events and choices—in the household lifecycle in this model. Time is discrete, and one period corresponds to four years in this model. Let j denote the agents’ age. They live with their parents until they graduate from high school at the beginning of age $j = J_E(= 20)$; until then, they make no decisions. After graduating high school, they choose whether to attend college or enter the labor market after graduating high school, represented as a node “High-School Graduation” in the figure. If they do not enroll in college, they enter the labor market as a high school graduate. If they choose to attend college, it takes four years (one model periods) to complete, and they enter the labor market as a college graduate after graduation, represented as a node “College Graduation” in the figure. This model does not consider the possibility of college dropout, as the dropout rate is negligible in Japan.¹²

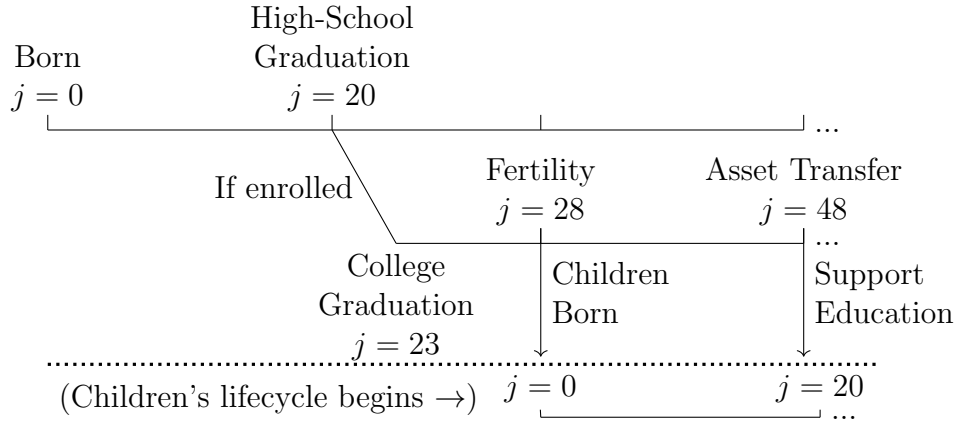


Fig 1: Key stages in the model’s lifecycle.

After completing their education, they enter the labor market. At the beginning of age $j = J_F(= 28)$, they make fertility choices by choosing how many children they have, $n \in \{0, 1, 2, 3, 4\}$.¹³ The timing of this decision is the same for both high school and college graduates. The lifecycle of a new cohort starts at this point (provided that the fertility rate is positive), represented in the bottom half of Fig 1. After their children (if they have any) graduate from high school, corresponding to the beginning of the age $j = J_{IVT}(= 48)$ for parents, they decide how much money to transfer to their children.

¹²For example, the dropout rate was 2.5% in 2021, according to the Ministry of Education, Culture, Sports, Science and Technology.

¹³I discuss this fixed fertility timing assumption in Appendix C.(iii).

This inter vivos transfer (IVT) decision affects the children’s college enrollment choices. Households retire from the labor market at the beginning of age $j = J_R(= 64)$. After that period, they face mortality risks; every period, a certain fraction of them is hit by exogenous mortality shocks and exits from the economy. They can live for J ($:= 103$) years at the longest, and they exit the economy after age J .

Dividing the model’s lifecycle into several stages, summarized in Table 1, helps us understand its structure; according to these stages, I formulate household problems in Section 4.3. Also, to provide a whole picture of the model’s lifecycle, I provide lists of choice and state variables in Tables 2 and 3 in advance, including their relevant stages. Each variable is explained in detail in the following subsections.

Stages	Corresponding age	
	High-School Graduates	College Graduates
(1) Education stage	20	20–23
(2) Early working stage (before fertility stage)	20–27	24–27
(3) Fertility stage	28	28
(4) Middle working stage (with children)	28–47	28–47
(5) Inter vivos transfer stage	48	48
(6) Late working stage (after children leave home)	49–63	49–63
(7) Retirement stage	64–	64–

Table 1: The lifecycle stages and the corresponding age. *Note:* Numbers in the second and third columns indicate the corresponding age for high-school and college graduates, respectively.

Notation	Description	Age/Period
<i>Throughout the entire stages</i>		
$c > 0$	Consumption	20–
$l \in [0, 1]$	Leisure	20–
$a' \in [\underline{A}, \infty)$	Asset	20–
<i>Education stage</i>		
$e \in \{HS, CL\}$	Education: high-school ($=HS$) or college ($=CL$)	20
<i>Fertility, Middle working, & Inter vivos transfer stages</i>		
$n \in \{0, 1, 2, 3, 4\}$	The number of children (fertility)	28
$q > 0$	Expenditures on children	28 – 47
$a_{IVT} \geq 0$	Inter vivos transfer	48

Table 2: Summary of the choice variables over the life stages. *Note:* The borrowing limit, $\underline{A}$, varies over the lifecycle, as discussed in the following subsections.

Notation	Description
<i>Throughout the entire stages</i>	
j	Age
a	Asset
<i>Education stage</i>	
ϕ	Psychic costs of college enrollment
I	Household income
<i>From early working stage to retirement</i>	
e	Educational background: high-school or college
z	Labor productivity
h	Innate ability
<i>Middle working stage</i>	
n	The number of children
<i>Inter vivos transfer stage</i>	
h_k	Children's innate ability
ϕ_k	Children's psychic costs of college education

Table 3: Summary of the state variables over the life stages.

4.2 Economic Environment

Demographics: Every period, a mass of new cohorts enters the economy. The size of the new cohort is endogenously determined according to fertility choices of cohorts in the fertility stage. The age distribution of this economy is thus determined by (1) households' fertility choices, which are endogenous, and (2) mortality risks after retirement, which are exogenous. Let $\zeta_{j,j+1}$ denote the probability of surviving at age $j + 1$ conditional on surviving at age j for each $j \in \{J_R, \dots, J\}$ with $\zeta_{J,J+1} = 0$.

Intergenerational Linkages and Initial Endowments: After graduating high school (at the beginning of age 20), agents are endowed with a triple (a_{IVT}, h, ϕ) : (1) assets transferred from their parents (a_{IVT}), (2) innate ability (h), which governs education returns in future earnings, and (3) psychic costs of college education (ϕ). They draw the innate ability from a distribution $g_{h_p}^h$, varying with the parents' innate ability level h_p . They also draw the psychic costs from a distribution g_{h,e_p}^ϕ , varying with the student's innate ability h and parent's education e_p .

Preferences: Every period, they draw utility from consumption c and leisure l according to a utility function $u(c, l)$. They discount future utility by a factor β . At age $j = J_F$, they choose the number of children they have, denoted by $n \in \{0, 1, 2, 3, 4\}$. They draw

additional utility from having children in several ways. First, they draw utility from expenditures q per child until the children graduate from high school, captured by a utility function $v(q)$. Note that this expenditure does not affect the child's state variables that are taken into account when education choices are made. Therefore, the function $v(q)$ can essentially be interpreted as another type of consumption good that parents enjoy. The utility $v(q)$ is discounted by a function $b(n)$, increasing and concave in the number of children n . Therefore, having more children (i.e., a higher n) also increases utility, holding other factors constant. Further, based on altruistic motives, parents draw utility by transferring assets to their children after the children graduate from high school. More specifically, parents consider children's expected lifetime utility in their education stage, with a discount rate $\lambda_a \cdot b(n)$, where λ_a represents the altruistic discount factor.

Costs of Children: Having children entails both monetary and time costs. First, raising children entails a monetary cost of $n \cdot q$, where n and q represent the number of children and the per-child expenditure, both chosen by the household. Also, they will make an additional expenditure $n \cdot a_{IVT}$ upon high school graduation of their children.¹⁴ In addition to the monetary costs, having a child requires κ units of time until the child graduates from high school and becomes independent.

Labor Earnings: Households choose hours worked to earn income. The labor earnings of a household are given by $w_e \eta_{j,z,e,h}(T - l)$; it is a product of the market wage w_e , labor productivity $\eta_{j,z,e,h}$, and hours worked $(T - l)$. The market wage w_e depends on the education level $e \in \{HS, CL\}$, where $e = HS$ and $e = CL$ represent high school and college graduates. Here, T denotes the disposable time that can be devoted to work or leisure where $T = 1 - \kappa \cdot n$ if they have n children who have not graduated high school and $T = 1$ otherwise. Labor productivity depends on age j , idiosyncratic component z , education level e , and innate ability h . The idiosyncratic component z follows a Markov process $\pi_z(z, z^-)$, where z^- denotes the component in the previous period.

Financial Markets: Financial markets are incomplete due to the lack of state-contingent claims. Households can self-insure against negative labor productivity shocks by savings, accruing interest payments at a rate of r . Households in the working stages can borrow at rate $r^- = r + \iota$ where $\iota > 0$ (i.e., borrowers incur the overseeing costs ι) up to a borrowing limit $\underline{A}$. In contrast, retired households are not allowed to borrow, which is a common assumption in the literature. In addition, eligible students have access to student loans subsidized by the government, which entails the interest rate of $r^s = r + \iota_s$. This loan is income- and ability-tested, and eligible students can borrow up to a limit $\underline{A}_s$.

Production Technology A representative firm chooses labor and capital inputs in competitive factor markets to produce final goods. There are two types of labor inputs in this economy; the college graduates (skilled) and high school graduates (unskilled). Their total labor supply in efficiency units are represented as L_{CL} and L_{HS} , respectively. I allow

¹⁴Children within a household are homogeneous, as assumed in the literature (e.g., [Abbott, Gallipoli, Meghir, and Violante, 2019](#)). Thus, education spending and IVT do not vary among children within the household.

them to be imperfect substitutes by considering the aggregate labor in efficiency units, L , is given as:

$$L = [\omega_{HS} \cdot (L_{HS})^\chi + \omega_{CL} \cdot (L_{CL})^\chi]^{1/\chi},$$

where $\omega_{HS} \equiv 1 - \omega_{CL}$ and $\omega_{CL} \in [0, 1]$ govern the relative productivity of the skilled workers, and χ governs the elasticity of substitution between the skilled and unskilled workers. The representative firm operates with a Cobb-Douglas production function with aggregate capital K and labor L to produce the output Y :

$$Y = ZK^\alpha L^{1-\alpha},$$

where Z represents the factor neutral productivity. Capital depreciates at the rate of δ , and the firm has to incur the capital depreciation costs.

Government The government raises the revenue by levying three types of taxes: consumption, labor income, and capital income taxes, where each tax rate is represented as τ_c , τ_w , and τ_a . In addition, the government collects accidental bequests and devotes them to cover expenditures. They use this revenue to fund (1) the public pension benefits, which gives p units of money to retired households each period, (2) subsidized loans for college students, for which the borrowing limit for a student with household income I is denoted by $\underline{A}_s(I)$, (3) grants for eligible college students, where the payment per eligible student is represented by $\bar{g}(I)$ where I represents household income when the student is in the education stage (i.e., their parent's age is J_{IVT}), (4) other public expenditures on college education, (5) lump-sum transfers ψ that is introduced for replicating the progressivity of labor income tax schedule in a simple way following the literature, (6) cash benefits for households with children under 19 with per-child payment of B , and (7) other expenditures S . The government budget constraint is given as follows:

$$\begin{aligned} \tau_c \cdot C + \tau_w \cdot (w_{HS}L_{HS} + w_{CL}L_{CL}) + \tau_a \cdot r \int_{\mathbf{x}} \mathbb{I}_{a(\mathbf{x}) > 0} a(\mathbf{x}) dF(\mathbf{x}) + Q \\ = p \cdot \mu_{old} + \psi + B \cdot \mu_{j \leq 19} + S + \underbrace{(\iota - \iota_s) \cdot K_s + G + \mu_s \cdot A_s}_{\text{expenditures on college education}}, \end{aligned} \quad (4)$$

where C , Q , μ_{old} , $\mu_{j \leq 19}$, K_s , G , μ_s , and A_s represent the total consumption, total accidental bequests, population mass of retired households, that of children under age 19, the total amount of borrowing by college students, the total grant payments, the population mass of college students, and other public expenditures on college education per student. The term $\tau_a \cdot r \int_{\mathbf{x}} \mathbb{I}_{a(\mathbf{x}) > 0} a(\mathbf{x}) dF(\mathbf{x})$ represents revenue from capital income taxation, where $\mathbf{x}$, $a(\mathbf{x})$, $\mathbb{I}_{a(\mathbf{x}) > 0}$, and $F(\mathbf{x})$ represent the household state vector, household's policy function for savings, an indicator function returning one if $a(\mathbf{x})$ is positive, and distribution over the state space.

4.3 Household Problems

Building on the primitives described so far, this section discusses and formulates household problems at each stage, as summarized in Table 1. The first is the education stage in which they choose whether to attend college or enter the labor market after graduating

high school.

(1) Education Stage: The state variables for the students are comprised of asset a_{IVT} , the innate ability h , psychic costs ϕ , and their parent's income I . They compare the expected value for entering the labor market as a high school graduate, $\mathbb{E}V^w$, with the value for enrolling in college V_{g1} , which is the expected lifetime utility conditional on enrolling in college, net of the psychic cost ϕ . They choose college enrollment if the latter is greater than the former; otherwise, they enter the labor market. The decision problem for college enrollment, along with the corresponding value function V_{g0} , is formulated as follows:

$$V_{g0}(a_{IVT}, \phi, h, I) = \max_{e \in \{HS, CL\}} \left\{ (1 - \mathbb{I}_{e=CL}) \cdot \underbrace{\mathbb{E}_{z_0} [V^w(a_{IVT}, j = 20, z_0; e = HS, h)]}_{\text{Lifetime utility for high school graduates}} + \mathbb{I}_{e=CL} \cdot \underbrace{[V_{g1}(a_{IVT}; h, I) - \phi]}_{\text{Value of attending college}} \right\}, \quad (5)$$

where $e \in \{HS, CL\}$ indicates the education choice: $e = CL$ means college enrollment and $e = HS$ does entering the labor market as a high school graduate. The indicator $\mathbb{I}_{e=CL}$ takes the value 1 if $e = CL$ and 0 otherwise. V^w denotes the value function for workers, representing the lifetime utility of entering the labor market immediately after high school graduation. The formulation of V^w is provided in Appendix C.(i). The initial draw of z , z_0 , is uncertain and is according to the invariant distribution of z .

V_{g1} in the value function (5) represents the gross value of attending college, not accounting for the psychic costs ϕ . Proceeding to college will increase their lifetime earnings, while this return from education depends on innate ability. In addition to the psychic costs, there are other two types of costs of proceeding to college. First, they have to pay tuition fees p_{CL} while enrolling in college. Second, although they can supply labor and earn some income, they have to spend $\bar{t}$ fraction of their time on study, which is the opportunity costs of going to college. They can fund their consumption and tuition fees through (1) transfers from their parents a_{IVT} , (2) labor earnings by themselves, where the hourly wage is given by the market wage for high-school graduates, w_{HS} , (3) borrowing through student loans if eligible, (4) government-provided grants if eligible. The problem for college students and the associated value of attending college V_{g1} are formulated as follows:

$$\begin{aligned} V_{g1}(a_{IVT}; h, I) &= \max_{c, l, a'} \{u(c, l) + \beta \mathbb{E}_{z_0} [V^w(a', j = 23, z_0; e = CL, h)]\} \\ \text{s.t. } & (1 + \tau_c)c + p_{CL} + a' = a_{IVT} + \bar{g}(I) + (1 - \tau_w)w_{HS}(1 - \bar{t} - l) + \psi, \\ & c > 0, \quad l \in [0, 1 - \bar{t}], \quad a' \geq -\underline{A}_s(I). \end{aligned}$$

Recall that ψ denotes the lump-sum transfer, the borrowing limit depends on household income I , and $\bar{g}(I)$ denotes the grant (cash transfer) provided by the government.

(2) Early Working Stage: The household problem in this period ends up with a standard lifecycle problem in which households choose consumption-saving and leisure-labor decisions. The value function is formulated in Appendix C.(i).

(3) Fertility Stage: When they reach the age of $j = J_F$, they choose how many children they have. The state variables in this stage consist of asset position a , productivity z , education e , and innate ability h . The problem is to choose the number of children n that maximizes expected utility from this point onward, with the value function formulated as follows:

$$V^f(a, z, e, h) = \max_{n \in \{0,1,2,3,4\}} \left\{ V^{wf}(a, j = J_F, z; e, h, n) \right\},$$

where $V^{wf}(a, j, z; e, h, n)$ in the parenthesis represents the value function for the middle working stage with n children represented below.

(4) Middle Working Stage (with children if applicable): The state variables in this stage consist of asset position a , age j , productivity z , education e , innate ability h , and the number of children n . The value function V^{wf} , after fertility choices until their children become independent (i.e., for age $j \in \{J_F, \dots, J_{IVT} - 1\}$), is formulated as follows:

$$\begin{aligned} V^{wf}(a, j, z; e, h, n) = & \max_{c, l, q, a'} \{ u(c/\Lambda(n), l) + b(n) \cdot v(q) \\ & + \left\{ \begin{array}{ll} \beta \mathbb{E}_{z'} [V^{wf}(a', j+1, z'; e, h, n)] & \text{if } j \in \{J_F, \dots, J_{IVT} - 2\} \\ \beta \mathbb{E}_{z', \phi_k, h_k} [V^{IVT}(a', z'; \phi_k, h_k, e, h, n)] & \text{if } j = J_{IVT} - 1 \end{array} \right\} \\ \text{s.t.} & \\ & (1 + \tau_c)(c + nq) + a' = Y_{wf}, \\ & a' \geq -\underline{A}, z' \sim \pi(z', z), h_k \sim g_h^{h_k}, \phi_k \sim g_{e, h_k}^{\phi_k}, \end{aligned}$$

Here, ϕ_k and h_k denote the psychic costs of college education and innate ability for their children. The household consumption is deflated by the equivalence scale $\Lambda(n)$, depending on the number of children n . The expenditures on children, nq , enter the budget constraint, and utility from children, $b(n) \cdot v(q)$, enters the objective function in the periods with children. V^{IVT} represents the value function for the inter vivos transfer (IVT) stage at $j = J_{IVT}$, which is described in the following paragraph. Households are uncertain about their children's innate ability h_k and psychic costs ϕ_k until the beginning of age $j = J_{IVT}$. Note that the implicit assumption here is that the expenditures q do not affect children's characteristics (ϕ_k, h_k) .¹⁵

The term Y_{wf} in the budget constraint denotes the disposable income, which is given by:

$$Y_{wf} \equiv (1 - \tau_w)w_e \eta_{j, z, e, h} (1 - l - \kappa \cdot n) + n \cdot B + \psi + a_{wf}(a)$$

Recall that B denotes the per-child cash transfer provided by the government. The term $a_{wf}(a)$ in the disposable income Y_{wf} represents the asset position at beginning of this stage, expressed as a function of the asset position at the end of the previous period, a . If a is negative (i.e., the household borrows), it has to pay the interest rate r^- . If a is

¹⁵I discuss this assumption in Appendix C.(iii).

positive (i.e., the household saves), it earns the interest rate r net of the capital tax τ_a . Thus, the asset position at the beginning of this period is given by:

$$a_{wf}(a) = \begin{cases} (1 + (1 - \tau_a)r)a & \text{if } a \geq 0 \\ (1 + r^-)a & \text{otherwise} \end{cases}$$

(5) Inter Vivos Transfers Stage: At the beginning of age $j = J_{IVT}$, households decide how much to transfer to their children. This timing corresponds to when the children graduate from high school and face the college enrollment choice. Two characteristics of their children are realized at this point: their psychic costs ϕ_k and innate abilities h_k . After observing the realized characteristics, households choose the optimal amount of per-child transfer a_{IVT} , formulated as follows:

$$V^{IVT}(a, z; \phi_k, h_k, e, h, n) = \max_{a_{IVT} \geq 0} \left\{ V^w(a - \tilde{a}_{IVT}, j = J_{IVT}, z; e, h) + b(n) \cdot \lambda_a \cdot V_{g0}(a_{IVT}, \phi_k, h_k, I) \right\},$$

where $\tilde{a}_{IVT} = \frac{n \cdot a_{IVT}}{1 + (1 - \tau_a)r}$ and the parent's state vector pins down household income I . Driven by altruism, they care about the lifetime utility of each child— $V_{g0}(a_{IVT}, \phi_k, h_k, I)$, as defined in equation (5)—which they discount by an altruistic factor λ_a .

(6) Late Working Stage: At this stage for households aged $j \in \{J_{IVT}, \dots, J_R - 1\}$, the household problem is similar to the early working stage: they make consumption-saving and leisure-labor decisions until retirement. The value function is formulated in Appendix C.(i).

(7) Retirement Stage: At the beginning of age J_R , households are forced to retire from the labor market. After that, they spend all their time on leisure and make consumption-saving decisions. Two additional aspects distinguish this stage from earlier choice problems: they receive the pension benefit p from the government and face uncertainty about the next period's survival. The value function is formulated in Appendix C.(i).

4.4 Stationary Equilibrium

I solve the stationary equilibrium of this economy. In equilibrium, households make every choice to maximize their expected utility, the firm maximizes its profit, and the government budget is balanced. Stationarity implies that the distribution over state variables is invariant. Importantly, the age distribution is determined endogenously according to households' fertility choices. See Appendix C.(ii) for the detailed definition of equilibrium and Appendix D.(i) for the computational algorithm for solving the equilibrium.

5 Calibration

To calibrate the model, I use the Japanese Panel Survey of Consumers (JPSC), a panel survey of Japanese women and their household members. It started in 1993 with a

representative sample of 1,500 women aged 24 – 34 and contains information about, for example, their income, educational background, marital status, fertility, and expenditures to detailed categories, including those on children’s education. I focus on the cohort born in 1959-69, the oldest cohort of this survey, especially to compute the completed fertility and intergenerational mobility of education. I keep only married households as in previous works (e.g., [Daruich and Kozlowski, 2020](#)) because the model focuses on choices made within married households such as fertility and educational investments.¹⁶ Unless otherwise mentioned, targeted moments described below are computed based on the JPSC’s 1959-69 cohort data.

Calibration strategy: As is standard, I follow a two-step calibration procedure. First, I set some parameters based on the literature or estimate them outside of the model. Then, the remaining parameters are calibrated internally to minimize the distance between the targeted moments observed in the data and those implied by the model. Let $\mathbf{d}$ be a vector of n targeted moments observed in the data, and let $\mathbf{m}(\boldsymbol{\theta})$ be the corresponding vector of model-implied moments, where the parameter vector $\boldsymbol{\theta}$ is of dimension $k \leq n$. I determine $\boldsymbol{\theta}$ by minimizing the following objective function:

$$\min_{\boldsymbol{\theta}} [\mathbf{d} - \mathbf{m}(\boldsymbol{\theta})]' [\mathbf{d} - \mathbf{m}(\boldsymbol{\theta})]$$

I determine 20 parameters endogenously, targeting 45 moments (i.e., $n = 45$ and $k = 20$). In what follows, I describe the functional forms and discuss which empirical moments are informative in identifying each parameter, organized by category (e.g., preferences and technology). Tables 11 and 12 in Appendix D.(ii).(C) summarize the parameters determined externally and those calibrated internally. Table 5 reports the model-implied moments alongside their data counterparts.

Preferences: Instantaneous utility for households are given as follows:

$$u(c, l) = \frac{(c^\mu l^{1-\mu})^{1-\gamma}}{1-\gamma}.$$

μ is internally determined as 0.23 so that the households spend one-third of the total disposable time on market work. Instantaneous utility from expenditures on a child is given as:

$$v(q) = \lambda_q q^{\gamma_q},$$

where λ_q is set to 0.62 so that the annual educational expenditure per child amounts to 7% of average income at age 28. The utility must be always positive (or always negative) in models of altruism with endogenous fertility, and I set $\gamma = 0.5$ following the literature (e.g., [Daruich and Kozlowski, 2020](#)). The altruistic discount factor λ_a is set to 1.03 so that the annual average IVT for college students amounts to 27% of average income at age

¹⁶Hence, an agent or household in this model refers to households with two individuals. Then, “children” in this model can also be interpreted as a household unit. That is, having n children can be interpreted as reproducing $n/2$ units of households. For example, if a household has two children, it means reproducing one household unit with two individuals.

28.¹⁷ Following [Darulich and Kozłowski \(2020\)](#), I set the discount function of the number of children $b(n)$ be as follows:

$$b(n) = \lambda_n n^{\gamma_n}$$

non-parametric, and assume that $b(n) = b_n$ for each $n \in \{0, 1, 2, 3, 4\}$ with $b(0) = b_0 = 0$. Those parameters are determined so that the model replicates the distribution of the completed fertility. The time for studying $\bar{t}$ is set to 0.8 so that the income share of labor earnings for college students in the model is close to 21% (SLS, 2018). I assume that $\beta = 0.98$ as in [Zhou \(2022\)](#).

Financial Markets: I set the borrowing limits $\underline{A} = 20$ million yen and $\underline{A}_s = 2.88$ million yen.¹⁸ $\underline{A}$ and $\underline{A}_s$ correspond to 3.6 and 0.52 relative to the average household income at age 28. The borrowing wedges are set $\iota = 0.055$ and $\iota_s = 0.054$ so that the model approximates the share of working households with a negative net worth (54%) and the share of students borrowing from the government-provided student loans (44%).

School Taste: I assume that the psychic costs ϕ are given as $\phi = \psi_{CL} \cdot \exp(-\nu \cdot h) \cdot \tilde{\phi}$. First, ψ_{CL} governs the scale of psychic costs and thus college enrollment rate. The second term $\exp(-\nu \cdot h)$ allows high ability students to have smaller psychic costs, as standard in the literature, and ν governs the education sorting by ability. Finally, $\tilde{\phi}$ is stochastic, and the distribution depends on their parent's education. Following [Darulich and Kozłowski \(2020\)](#), I assume that $\tilde{\phi}$ is distributed on an interval $[0, 1]$ and follows the following CDF:

$$G_{e^p}^{\tilde{\phi}} = \begin{cases} \tilde{\phi}^\omega & \text{if } e^p = 0 \\ 1 - (1 - \tilde{\phi})^\omega & \text{if } e^p = 1 \end{cases}$$

Here, ω governs the intergenerational transmission of school tastes. ψ_{CL} is set to 20.8 so that the model approximates the college enrollment rate of 37.7%. ν is set to 1 in the benchmark, and ω is set to 1.71 so that the intergenerational transition matrix of education in the model matches the data counterpart as closely as possible. Table 4 reports the transition matrix in the model and data. (i, j) -th entry of the matrix indicates the probability that children acquire skill (education) $j \in \{HS, CL\}$ given that their parent's skill is $i \in \{HS, CL\}$ in the benchmark model, and values in parentheses represent the data counterparts. The table indicates that the education level is persistent across generations: children of high school graduates attend college with a probability less than 0.3, whereas children of college graduates do with a probability approximately 0.6.

¹⁷Although some parents whose children do not enroll in college make IVT in the model, the amount is negligibly small. One reason is that the marginal gains from IVT are more significant if their children attend college, as students are financially constrained, primarily because of their limited earning ability.

¹⁸The former is based on the Family Income and Expenditure Survey (FIES) by the Ministry of Internal Affairs and Communications and the latter is based on the SLS (2018).

Parents/Children	High school	College
High school	0.725 (0.798)	0.275 (0.202)
College	0.412 (0.423)	0.588 (0.577)

Table 4: Intergenerational transition matrix of education. *Note:* (i, j) -th entry of the matrix indicates the probability that children acquire skill (education) $j \in \{HS, CL\}$ given that their parent’s skill is $i \in \{HS, CL\}$ in the benchmark model, and values in parentheses represent the data counterparts.

Intergenerational Transmission of Ability: The intergenerational transmission of ability is according to the following formula:

$$\begin{aligned}\log(h) &= \rho_h \cdot \log(h_p) + \varepsilon_h, \\ \varepsilon_h &\sim N(0, \sigma_h).\end{aligned}$$

I internally determine ρ_h to 0.30 to approximate the intergenerational income elasticity (0.3) and also set σ_h to 0.65 so that the variance of log income at age 28 in this model is close to 0.27.

Income Process and Education Return: The efficiency labor of an agent aged j , education e , innate ability h , and productivity z , $\eta_{j,z,e,h}$, is given as follows:

$$\begin{aligned}\log \eta_{j,z,e,h} &= \log[f^e(h)] + \gamma_{j,e} + z, \\ f^e(h) &= h + e \cdot (\alpha_{CL} h^{\beta_{CL}}), \\ z' &= \rho_e^z z + \zeta, \quad \zeta \sim N(0, \sigma_e^z).\end{aligned}$$

To set $\gamma_{j,e}$, I estimate the second-order polynomial of hourly wages on age using JPSC. As reported in Table 10 in Appendix D.(ii).(C), the income gradient on age is larger for college graduates than the rest of the workers, but the degree is modest compared with the US case (e.g., [Abbott, Gallipoli, Meghir, and Violante, 2019](#)).

Following [Darulich \(2022\)](#), I construct a series of four-year total household income (i.e., the sum of the husband’s and wife’s labor earnings) for each couple and estimate the stochastic component z of the income process separately by education group. The estimated persistence parameters (ρ_e^z) for high school and college graduates are 0.920 and 0.924, and the parameters governing income volatility (σ_e^z) are 0.060 and 0.048, respectively. These levels are similar to those estimated by [Darulich \(2022\)](#) using U.S. data with a four-year period, as in this study.¹⁹ The function $f^e(h)$ indicates that the education return depends on innate ability h : individuals with higher innate abilities can obtain greater returns through college education. Following [Darulich and Kozłowski \(2020\)](#), α_{CL} and β_{CL} are determined so that the model replicates the ratio of log wage between college graduates and the rest of the population at age $j = 28$ and the log wage variance for college graduates at age $j = 28$.

Production: I set the capital share $\alpha = 0.31$ based on JIP database (2008) by the Research Institute of Economy, Trade and Industry (RIETI) in Japan. I set $\delta = 0.07$ following [Kitao \(2015\)](#). χ is set to 0.70, following the cross-country estimates of [Taniguchi](#)

¹⁹The details of the estimation procedure can be found in Appendix D.(ii).

and Yamada (2024). ω_{CL} is internally determined to 0.52 so that the average wage ratio between college graduates and the rest in the model amounts to 1.36. Z is determined so that the wage rate for high school graduates is normalized to one in the benchmark.

Government: Tax rates are set to $\tau_c = 0.1$, $\tau_w = 0.35$, and $\tau_a = 0.35$ in the benchmark. The lump-sum transfer is set to $\psi = 0.01$ to match the ratio between the variance of log net income and that of log gross income (0.6). The pension benefit p is set so that the government provides ¥70,000 per person per month (i.e., $p = ¥70,000 \times 2$ persons), consistent with the amount reported by Japan Pension Service (JPS).²⁰ The monthly payment of ¥70,000 for two persons amount, annually, to 31% of the average household income at age 28. The cash transfer B , which we refer to “child benefit” or “typical pro-natal transfers” hereafter, is given as ¥10,000 per child per month, approximating the actual payment.²¹ The monthly payment of ¥10,000 amount annually to 2% of the average household income at age 28. The other expenditure S is set so that the government budget constraint is balanced in the benchmark and fixed throughout the counterfactual experiments. I internally calibrate A_s to match government expenditure on college education as a share of GDP, which is 0.3%, in the benchmark model.²² This corresponds to about 1.8% of total government expenditure in both the model and the data, a small amount that is consistent with the fact that the Japanese government provides relatively little subsidy to college education compared with other countries, as discussed in Appendix A.

Miscellaneous: The survival probability $\zeta_{j,j+1}$ is set based on the Vital Statistics (2019).²³ Annual college tuition fees p_{CL} are set to 1.05 million yen. This corresponds to 20% of the household income at age 28. κ is set to 0.044 so that households spend 13.3% of their working hours on childcare, consistent with data from the Survey on Time Use and Leisure Activities (STULA).

6 Validation

This section validates the calibrated model, particularly its fertility behavior, which is crucial for confidence in the quantitative analysis. I demonstrate that the benchmark model successfully replicates several moments not targeted in calibration: (1) fertility differentials between college and high school graduates, (2) fertility elasticity with respect to pro-natal cash transfers, (3) cross-country estimates of fertility elasticity with respect to college subsidies, and (4) the composition of students’ income among parental transfers (IVT), loans, and labor earnings.

²⁰See, <https://www.nenkin.go.jp/international/japanese-system/nationalpension/nationalpension.html>.

²¹See, for example, the website by the Ministry of Health, Labour and Welfare (MHLW), <https://www.mhlw.go.jp/english/topics/child-support/>.

²²For data on public expenditures on tertiary education, see https://www.mext.go.jp/b_menu/shingi/chukyo/chukyo0/toushin/attach/1335651.htm (in Japanese).

²³See, <https://www.mhlw.go.jp/english/database/db-hw/outline/index.html>.

6.1 The fertility differential across education levels

More educated parents have fewer children than less educated parents. According to my JPSC sample, college graduates had a completed fertility rate of 1.92, lower than 2.12 for high school graduates, resulting in a college-to-high-school fertility ratio of 0.91.²⁴ In this benchmark model, the completed fertility of college graduate wives is 1.79, which is lower than the high school graduate wives', 2.28, resulting in a college-to-high-school fertility ratio of 0.80. The benchmark model captures the fertility differential in that more educated parents have fewer children than less educated parents.

Why does this fertility differential result in this model? First, the opportunity costs of having children are more significant for college graduates, making child-bearing more costly for them. Second, college graduate parents are more willing to pay for the IVT, making a child more costly. This is because the children's innate abilities and psychic costs of education, (h_k, ϕ_k) , govern education returns and the willingness to attend college, which in turn govern the marginal gains from IVT for parents.

Appendix B.(i) presents simple static models that highlight these mechanisms in closed forms. These two mechanisms align with the major explanations proposed by the theoretical literature on the fertility differentials across income distribution.²⁵ (1) having children is more costly for higher-income households, provided that having children is time-intensive, and (2) higher-income households have higher demands for child quality or an advantage in parental investments.

6.2 The benefit elasticity of fertility

Previous studies show that cash benefits such as the child benefit and baby bonus have a significant impact on fertility. Many of them report that the benefit elasticity of fertility, the percentage increase in fertility rate against the one percent increase in the cash transfer, is about 0.1 – 0.2. For example, Milligan (2005) studies a reform of Quebec's baby bonus and shows that an extra 1,000 Canadian dollars benefit would increase fertility by 16.9%, which implies a benefit elasticity of 0.107. Cohen et al. (2013) uses a variation in the child subsidy payment observed around 2003 in Israel, providing a larger subsidy for third or higher births. They show that the benefit elasticity of fertility was 0.176.²⁶

To examine how this model performs in this respect, I conduct the following exercise. Let B_0 denote the per-child cash transfers for households with children under 17 in the benchmark. I solve the household problem, holding prices, tax rate, and distribution fixed, with several levels of the per-child payment $B = B_0 \cdot (1 + x)$ for some $x \in X$,

²⁴This is also observed in the National Fertility Survey (NFS) provided by the National Institute of Population and Social Security Research (IPSS), another data source where we can check the completed fertility rate by different education levels. See, https://www.ipss.go.jp/site-ad/index_english/survey-e.asp. According to the NFS (2015), college graduate wives' completed fertility has been lower than less educated ones almost every survey year since 1977. The latest survey in 2015 reports that the completed fertility of wives with a college degree was 1.89, and that of high school graduate wives was 1.98.

²⁵See, for example, Jones, Schoonbroodt, and Tertilt (2010).

²⁶Also, González (2013) adopts the regression discontinuity design to study the effects of Spain's reform in 2007, introducing a one-time payment of 2,500 euros (about 3,800 USD) for births, almost 4.5 times the monthly minimum wage for full-time workers. It finds a statistically significant impact on fertility, increasing conceptions by 5 – 6%.

where I set $X = \{0.1, 0.2, \dots, 1.9, 2.0\}$, a reasonable range in the context of the expansion examined in the empirical studies. This procedure yields the implied fertility rate, and given the expansion rate x , I compute the corresponding benefit elasticity of fertility. Finally, I take the average of those 20 values for the implied elasticity and find that the average elasticity is 0.138 (with a standard deviation of 0.025), which is consistent with the empirical estimates.

6.3 The fertility–subsidy gradient

The model also replicates the cross-country estimate of the relationship between total fertility rates and the “generosity” of the higher education subsidy measured by the government expenditures on higher education per student relative to the country’s GDP per capita. I use cross-country panel data from the World Bank,²⁷ collecting more than 200 countries/regions over more than 50 years. I then regress the log of total fertility rates of a country i in year t on the generosity of the subsidy for the same country-year, controlling for country-fixed effects, time-fixed effects, and other covariates such as GDP per capita and demographic factors. I refer to the estimated coefficient for subsidy generosity as *the fertility–subsidy gradient*, which represents the percentage increase in total fertility associated with a one percentage point rise in government spending on higher education per student relative to GDP per capita.

Based on the sample of the OECD countries, the estimated coefficient is 0.0013, which is marginally significant (at 10% level). Based on the sample of the entire set of (more than 200) countries/regions, the estimated coefficient is 0.00027, which is statistically significant at the 1% level. I simulate an introduction of a uniform subsidy using the benchmark model by providing uniform cash transfers—that amount to 1% of the GDP per capita—to all college students. This simulation leads to a 0.11% increase in total fertility, which aligns with the estimate based on the OECD countries. The details of the regression analysis are provided in Appendix D.(ii).(B).

6.4 Composition of students’ income

Capturing the composition of students’ income—how college students finance their living expenses and tuition fees—is also important as it is critical not only to students’ education choices but also to parents’ IVT choices and, thereby, fertility choices. According to the Student Life Survey (2018), students’ income consists of three components. The largest share, 61%, comes from parental asset transfers. Labor earnings account for 21%, while the remaining 18% is financed by student loans. Although the 21% share of labor earnings is a targeted moment, the rest is not. Nevertheless, the model captures the overall income composition well: IVT and loans account for 66% and 14% of students’ income, closely matching the data counterparts.

²⁷See, <https://databank.worldbank.org/>.

Table 5: Targeted and non-targeted moments.

Moment	Data	Model
<i>Targeted</i>		
<i>College enrollment</i>		
College enrollment rate	0.38	0.41
Enrollment rate across income quintile	See top-right panel in Fig 2	
Intergenerational mobility		
Pr(Child=High-school Parent=High-school)	0.73	0.71
Pr(Child=College Parent=High-school)	0.27	0.29
Pr(Child=High-school Parent=College)	0.41	0.41
Pr(Child=College Parent=College)	0.59	0.59
<i>Income inequality & mobility</i>		
Skill premium	0.31	0.39
Var log wage for college graduates at age 28	0.14	0.13
Intergenerational income elasticity	0.30	0.35
Var(log disposable income)/Var(log gross income)	0.60	0.76
Income quintile at age 48	See top-left panel in Fig 2	
<i>Parental transfers</i>		
Average IVT for college students	0.40	0.37
IVT across income quintile	See bottom panel in Fig 2	
Average pre-college spending	0.07	0.07
Pre-college spending across income quintile	See middle-left panel in Fig 2	
<i>Fertility</i>		
Average completed fertility	2.04	2.00
Fertility across income quintile	See middle-right panel in Fig 2	
<i>Labor supply and saving</i>		
Work hours	0.33	0.32
Average income share of loans (students)	0.21	0.17
Capital-to-output raio (annual)	4.0	4.0
<i>Non-targeted</i>		
Fertility-subsidy gradient	0.13	0.11
Benefit elasticity of fertility	$\simeq 0.10$	0.138
Students' income composition (IVT:Loans:Labor earnings)	(0.61 : 0.18 : 0.21)	(0.70 : 0.13 : 0.17)

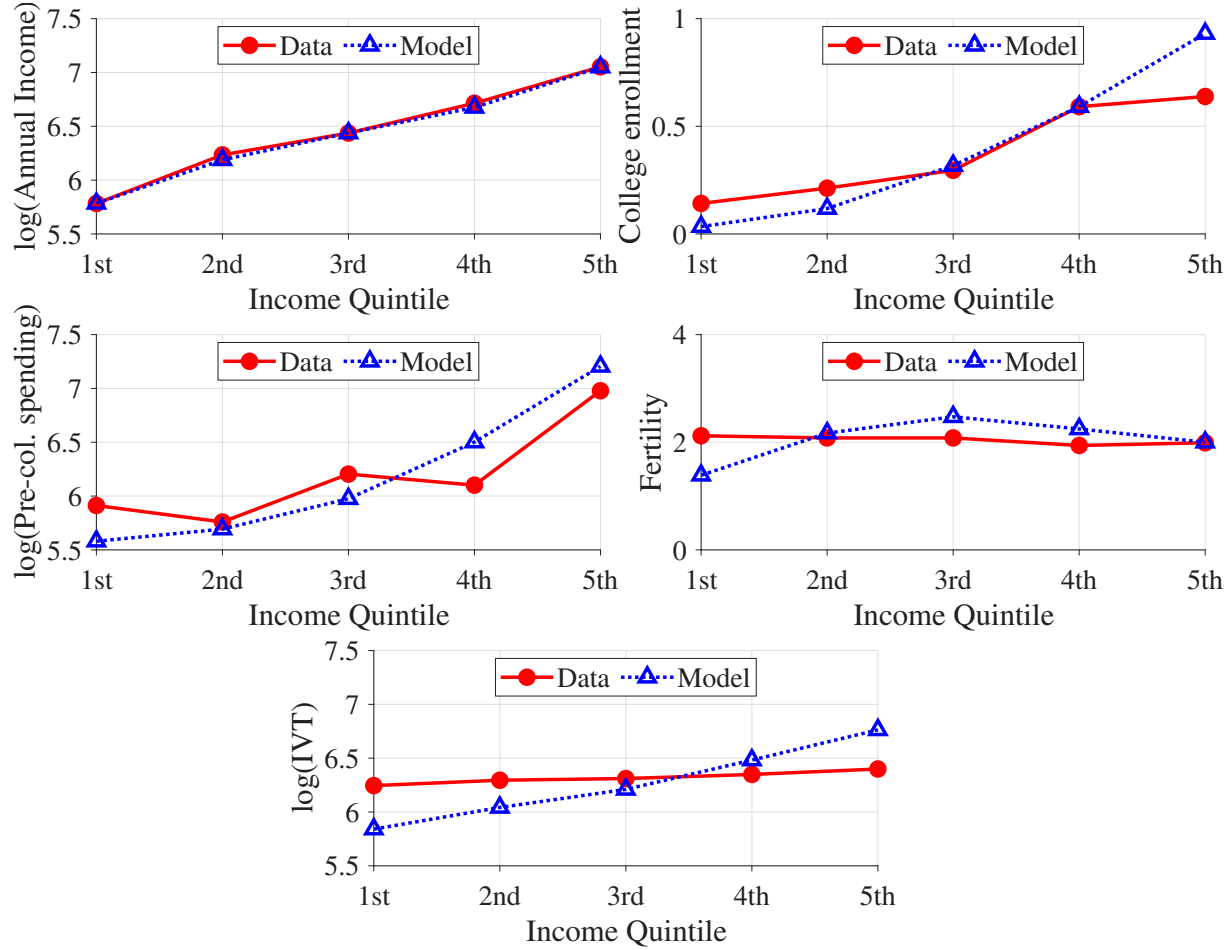


Fig 2: Key moments across income quintile.

7 Roles of an Income-tested College Subsidy

I introduce an income-tested college subsidy—which approximates the one actually introduced in Japan in 2020—into the calibrated model. More specifically, I allow for an income-tested subsidy for students in low-income households, whose income is below $\bar{I}$, and students in these households receive an amount g each period while they are in college. The grant function $\bar{g}(I)$ is formulated as follows:

$$\bar{g}(I) = \begin{cases} g & \text{if } I < \bar{I} \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

The income threshold $\bar{I}$ in the actual policy approximately corresponds to the 15th percentile of the household income distribution, and the payment amounts to approximately two-thirds of the average college student's expenses, including both tuition fees and living expenses.²⁸ I set $\bar{I}$ and g based on this information, considering the income distribution and students' expenditures in the benchmark model (initial steady state).

In the computation, I refer earnings potential ($w_e \eta_{j,z,e,h}$) instead of actual income

²⁸ Although the income threshold $\bar{I}$ and payment g in the actual system vary with certain characteristics of households and students, such as family structure and whether the student commutes to college from home, I abstract away from those details for simplicity.

(earnings potential multiplied by hours worked) to determine eligibility and the threshold for the subsidy $\bar{I}$. This approach rules out the possibility of households engaging in strategic behavior by reducing their labor supply to qualify for the subsidy.²⁹

Hereafter, I analyze the effect of the subsidy on fertility (Section 7.1), its long-run macroeconomic implications (Section 7.2), and the transitional dynamics (Section 7.3).

7.1 Effects on Fertility

The main goal here is to quantify the insurance effects of the income-tested subsidy discussed in Section 3. Therefore, I first simulate the introduction of the subsidy in a partial equilibrium setup, in which prices, tax rates, and distributions are fixed as in the benchmark, and see the (direct) effect on fertility.³⁰

I decompose the direct effects on fertility into two components: insurance effects and the *price effect*. The insurance effects are those discussed in Section 3. Although the illustrative model considers an income-tested subsidy with an expected value of zero, when the expected value is nonzero, as in the actual policy considered here, an additional effect arises: the price effect, which captures the reduction in the expected monetary costs (or “price”) of children.

In Appendix B.(ii).(C), I demonstrate that the direct effects can be decomposed into insurance and price effects as follows. First, I construct a grant function that provides the present discounted value of the expected college subsidy to each household at the fertility stage, based on the joint probability that its (potential) children enroll in college and that the household is eligible for the subsidy (i.e., its income will be below the income threshold). I then compute the effects of this subsidy on fertility, referred to as the price effect, which captures the impact on fertility resulting from the reduction in the expected cost of children. My decomposition formula indicates that the insurance effect can be computed as the difference between the overall direct effect and the price effect. Tables 6 and 14 summarize the decomposition results.

The main results of the decomposition are threefold. First, the insurance effects account for 62% of the direct effects on the total fertility rate (TFR) (i.e., 1.3 p.p. over a 2.1 % increase), while the price effects account for 38% (i.e., 0.8 p.p. over a 2.1% increase). This means that the insurance effect is quantitatively more important in accounting for the effect of the income-tested college subsidy on fertility. Second, the effects on fertility are more significant for college graduates (a 4.5% increase) than high school graduates (a 0.9% increase). This is because college graduates have a higher probability of being eligible, as their children are more likely to attend college, resulting in both stronger average and insurance effects for them compared to high school graduates. Third, the relative importance of insurance effects to price effects is more significant for high school graduates: the insurance effect accounts for about half of the fertility increase for college graduates, whereas it does about 90% of that for high school graduates. This is

²⁹Although such strategic behavior can occur in practice and be relevant for policy design, there are ways to mitigate this issue to some extent—for instance, by determining eligibility based on income from prior years. Therefore, for simplicity, I consider a setting that entirely rules out such behavior.

³⁰The effects on fertility in the long run, which also capture the indirect effects such as GE effect, taxation effect, and distributional effects, are reported in Appendix E.(ii). It turns out that the direct effects explain two-thirds of the overall effects.

because (1) on average, college graduates are more able to self-insure against income risk due to higher earnings, and (2) since children of high school graduates are less likely to attend college, the effect of reducing the expected cost (i.e., the price effect) is generally negligible.

	<i>Price</i>	<i>Insurance</i>	Total
Total fertility rate (Δ %)	+0.8	+1.3	+2.1
High school graduates (Δ %)	+0.1	+0.8	+0.9
College graduates (Δ %)	+2.4	+2.1	+4.5

Table 6: Direct effects on fertility and decomposition results.

Although the setup of the quantitative model differs from that of a simple model presented in Section 3, the interpretation of the insurance effect still applies. First, having one more child can be, ex-post, extremely costly in utility terms in two ways after the bad income realization. First, holding the amount of IVT constant, a bad income shock can lead to an extremely low consumption level for parents if they have more children. On the contrary, holding the household consumption level constant, a bad income shock may make it infeasible for them to spend enough IVT to let their children attend college if they have more children. Both situations are costly for parents, leading to a significantly high expected marginal cost of children. If the income-tested college subsidy is available, the expected marginal costs of children in utility terms become lower because they can receive some cash after facing a bad income shock and devote it to support their children’s education.³¹

Discussion: Applicability Beyond Japan It is natural to ask whether this mechanism applies beyond Japan. I argue that the main result—the insurance role of income-tested subsidies—extends to other high-income countries, although the effective implementation of such subsidies may differ across contexts. A condition that may be Japan-specific and central to the mechanism is that parents bear most of the financial burden of college education, and this may vary across countries. In some countries, students—not parents—finance college through loans, either because of cultural norms regarding parental responsibility or institutional features such as the availability and design of student loan programs.

Consider an extreme case in which parents do not contribute to college costs at all and students fully finance their education through borrowing. In such an environment, parents do not directly face the financial burden of future education costs when making fertility decisions. However, if parents themselves have outstanding student loans, that debt reduces their disposable income during child-rearing years. The resulting lower

³¹Appendix C.(iii) discusses the direction and magnitude of the potential bias introduced by the fixed fertility timing assumption in understanding this insurance mechanism. Also, Appendix E.(vi) examines other policies, including those insuring against income risks, such as (fertility-dependent) wage insurance, in an expenditure-neutral manner. I show that although the fertility-dependent wage insurance has a positive effect on fertility, the effect is smaller than the insurance effects of income-tested college subsidy. This is because the income-tested college subsidy provides partial insurance not only against income risk but also against another source of uncertainty: uncertainty regarding children’s characteristics (i.e., psychic costs and innate abilities) and the associated uncertainty regarding the willingness to pay for children’s education.

disposable income increases the risk of being unable to cover child-related expenditures, thereby discouraging fertility. This interaction between income levels, risk, and fertility is consistent with the quantitative results presented above. The insurance effect is particularly important relative to price effects for high-school graduates, whose incomes are relatively low.

Therefore, the insurance mechanism remains relevant even in economies where students finance higher education themselves, although it applies only to those who have completed college. In such settings, income-contingent loan programs—under which repayment obligations adjust to realized income—may operate similarly to income-tested subsidies by mitigating income risk and thereby providing insurance against costly states associated with having children.

7.2 Long-run Macroeconomic Implications

Next, I examine the long-run macroeconomic implications of the subsidy by solving the stationary equilibrium with the grant function $\bar{g}(I)$. The government adjusts the labor income tax rate τ_w to balance its budget. The main result here is that the effects of introducing the subsidy—such as a higher college enrollment rate, lower skill premium and income inequality, and increases in output and welfare—are quantitatively more significant if we take into account endogenous fertility.

I highlight the roles of endogenous fertility by solving the *exogenous fertility* version of the model, where the household decision rule (policy function) for fertility, $\bar{n}(a, z, e, h)$, are fixed as in the benchmark. Note that this exogenous fertility setup is common in the macroeconomic literature on college subsidies (e.g., [Krueger and Ludwig, 2016](#); [Matsuda and Mazur, 2022](#)). Comparing the results under exogenous fertility with those under endogenous fertility reveals the role of fertility choices in evaluating the macroeconomic implications of a college subsidy. Table 7 reports the main results under both endogenous and exogenous fertility.

	Fertility setup	
	Endogenous	Exogenous
College enrollment (Δ p.p.)	+4.0	+3.5
Skill premium (Δ %)	−8.3	−5.8
Standard deviation of log wage (Δ %)	−1.3	−1.1
Output per capita (Δ %)	+0.7	+0.5
Labor income tax (Δ p.p.)	+0.02	+0.13
Welfare (%)	+5.1	+3.5
Marginal Value of Public Funds (Willingness to pay / Net costs of policy)	302.94	17.06

Table 7: Long-run macroeconomic implications (endogenous vs. exogenous fertility). *Note:* For “Standard deviation of log wage,” wages for an agent with (j, z, e, h) is captured as $w_e \cdot \eta_{j,z,e,h}$. Welfare gains are represented in terms of consumption equivalence for newborn agents under the veil of ignorance.

First, the policy introduction results in a 3.5 p.p. increase in the college enrollment

rate under exogenous fertility, which is lower than a 4.0 p.p. increase under endogenous fertility. Under both exogenous and endogenous fertility, the grants enable some children who would otherwise be unable to afford college to enroll. Also, an increase in the college enrollment rate is amplified in the long run through a distributional effect: the composition of college graduates in younger generations increases, and when they reach parenthood, their children are more likely to enroll in college due to intergenerational linkages via assets, abilities, and tastes.³²

Under endogenous fertility, the college enrollment rate can further increase through an additional channel. As discussed in Section 7.1, the subsidy increases fertility particularly among college graduates, and their children are more likely to attend college due to intergenerational transmission of abilities and tastes. When these children become parents themselves, their own children, if they have any, are also likely to inherit similar traits, favoring college enrollment. In other words, fertility responses amplify the distributional effect on college enrollment. Through this fertility channel, the long-run share of college graduates increases further, explaining why the long-run college enrollment rate is higher under endogenous fertility.

A higher college enrollment rate implies a greater supply of college graduates in the labor market, leading to a reduction in the skill premium. The decline in the skill premium under endogenous fertility (-8.3%) is more significant than under exogenous fertility (-5.8%) due to higher college enrollment rates under endogenous fertility. When considering another proxy for income inequality, such as the standard deviation of log wages, the reduction in income inequality is also more significant under endogenous fertility (-1.3%) than under exogenous fertility (-1.1%), partly due to the lower skill premium.

The output per capita increases more under endogenous fertility ($+0.7\%$) than under exogenous fertility ($+0.5\%$), due to greater supply of college-graduate workers. The greater output implies the greater tax bases, leading to a lower tax rate in equilibrium under endogenous fertility (35.02%) than under exogenous fertility (35.13%). The welfare gain—measured as consumption equivalence for new born agents under the veil of ignorance—is greater under endogenous fertility ($+5.1\%$) than exogenous fertility ($+3.5\%$) because of those channels (i.e., a lower tax, lower inequality, and a higher expected lifetime income).³³

I also compute the average Marginal Value of Public Funds (MVPF), defined as the average willingness to pay for a policy divided by the aggregate (net) costs of the policy. The MVPF under endogenous fertility (302.94) is significantly greater than under exogenous fertility (17.06) because the policy is almost self-financing under endogenous fertility.³⁴

In Appendix E.(iii), I examine the expansion of the policy by increasing the income-threshold $\bar{I}$. The main results are threefold. First, the expansion of the policy further increases college graduates' fertility, and this fertility response contributes to a higher

³²A detailed decomposition exercise can be found in Appendix E.(ii).

³³In Appendix E.(i), I formulate this welfare measure and discuss theoretical and conceptual difficulties on welfare analysis with models with endogenous fertility.

³⁴Hendren and Sprung-Keyser (2020) examine the MVPFs for 133 historical policy changes over the past half-century in the U.S. and find that the MVPFs for college education subsidies were infinite in many cases, as they were self-financing. Krueger et al. (2024) also show that education policies, such as making college free and improving funding for public schooling, are more than self-financing using a GE-OLG model.

college enrollment rate via intergenerational linkages. Second, however, this expansion reduces the fertility of high school graduates because it significantly increases the intergenerational mobility of education and results in a “quantity-quality trade-off” of children. Third, the positive effects on output during expansion diminish and eventually turn negative as the threshold $\bar{I}$ becomes sufficiently high, since broader coverage significantly crowds out labor supply and savings among households with children.

7.3 Transitional Dynamics

Lastly, I investigate the transitional dynamics of the economy upon the introduction of the income-tested subsidy. The introduction occurs unexpectedly for households in the initial equilibrium (benchmark economy). Households have perfect foresight about future prices and tax rates and make subsequent choices to maximize utility. For computational reasons, I assume that the government collects the lump-sum taxes to balance the budget during the transition and that the labor income tax rate immediately reaches its final steady-state value, $\tau_w = 0.3502$.³⁵

The takeaway here is that it takes a long time for the macroeconomic effects to accrue, and that part of the reason is fertility responses are taken into account. For example, the left-panel of Fig 3 represents the transition path of college enrollment rates. College enrollment rates (within cohorts) increase upon the introduction but do not immediately reach the new equilibrium value and gradually converge to it. This is because a part of the long-run increase in college enrollment rate is explained by the distributional (composition) effects: students who benefit from the policy become parents in the future, whose children will be more likely to attend college due to intergenerational persistence of assets, abilities, and tastes. Also, the fertility responses strengthen the distributional effects: college graduates increase fertility, and those “marginal” children born are more likely to be college graduates, whose children will also be more likely to attend college. The roles of fertility margin can be observed in the difference between the transition path under endogenous fertility (a blue solid line) and that under exogenous fertility (a red dashed line). Both paths for college enrollment rates are nearly identical for the first 18 years after the policy introduction but begin to diverge thereafter. This divergence corresponds to the period when the first generation born after the policy starts making college enrollment choices.

³⁵Some previous studies also adopt this strategy for computational reasons (e.g., [Daruich, 2022](#)).

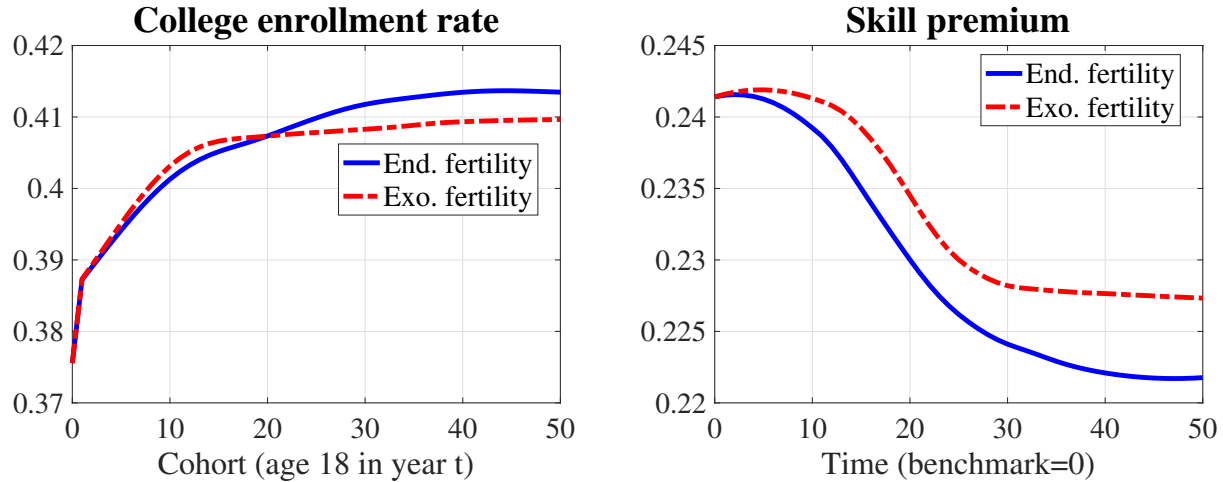


Fig 3: Transitional dynamics of college enrollment rates and skill premium. *Note:* “End. fertility” and “Exo. fertility” represent transition paths under endogenous fertility and exogenous fertility.

The right-panel of Fig 3 shows the transitional path of the skill premium. It decreases over time as college enrollment rates increase. Under exogenous fertility, the skill premium reaches its final value 30 periods (60 years) after the policy introduction, whereas under endogenous fertility, it takes more than 40 periods (80 years). This is because college enrollment rates take longer to reach their final steady-state level under endogenous fertility.³⁶

8 Concluding Remarks

This paper examines the macroeconomic implications of income-tested college subsidies, which are prevalent across countries, by constructing a lifecycle Aiyagari model with fertility and college enrollment choices. The main contributions of this paper are: (1) to analytically and quantitatively show that income-tested subsidies provide insurance for fertility choices under income uncertainty and (2) to show that the resulting fertility responses amplify the macroeconomic effects of the policy.

Although the model is applied to Japanese data, the underlying mechanisms, highlighted through both analytical and quantitative models, are not unique to Japan but offer insights applicable to other countries as well. Furthermore, while this study focuses on college education subsidies, the fertility insurance mechanism also applies to income-tested subsidies at earlier education levels and to a broader set of child-related transfers, although its quantitative importance and the resulting macroeconomic implications likely differ across policies. Exploring these broader policies would require extending the current model to incorporate human capital accumulation before college, which this study abstracts from.³⁷ This paper also abstracts competition for college admissions, which can be important to consider when evaluating broader education policies, such as education taxes and changing enrollment capacities.³⁸ Incorporating these factors into the current

³⁶Transitional paths for other variables, such as taxes, output, and welfare, are presented in Appendix E.(vii).

³⁷Examples of recent works in this respect include Lee and Seshadri (2019); Daruich (2022); Yum (2023); Moschini and Tran-Xuan (2023); Krueger, Ludwig, and Popova (2024).

³⁸See, for example, Kim, Tertilt, and Yum (2023); Kang (2024); Gu and Zhang (2024).

model and exploring broader child-related and education policies are left for future research.

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A Facts: Education Costs and Fertility Choices in Japan

Japan is a leading country in demographic aging,³⁹ leading to the shrinking labor force, output, and tax base, while the public expenditures on social security benefits are increasing. As a countermeasure against this demographic issue, the government introduced grants for college students in 2020. Two key underlying premises are: (1) it will increase the “quality” of the labor force in the long run, and (2) it will increase the fertility rate, which increase the “quantity” of the labor force in the long run. This pro-natal motive is explicitly described in an act for introducing the grants.⁴⁰

The aim is ... to foster an environment where people can bear and raise their children with a sense of ease by alleviating the economic burden associated with higher education, thereby contributing to addressing the rapid decline in the birthrate in our country.

(Act on Support for Higher Education Studies, enacted on May 17, 2019)

Although that expectation for education policies as a pro-natal policy is unconventional, there are facts suggesting that the financial costs for parents to support their children’s college enrollment are a significant barrier to fertility in Japan. I first list the five facts below and then elaborate on each one by one:

1. Japan is one of the least in subsidizing tertiary education.
2. Average parental transfers to college students amount to 65% of the average total education expenditures up to high school graduation.
3. About 80% of parents desire a college education for their children.
4. Couples stop at fewer children than their ideal number primarily due to financial costs.
5. Children’s college enrollment is associated with a higher likelihood of parents stopping at fewer children due to the financial reason.

Fact 1. *Japan is one of the least in subsidizing tertiary education.*

Japan is one of the least in subsidizing tertiary education, which roughly corresponds to four-year college education in Japan, given that the enrollment rates for tertiary education other than the four-year college are significantly lower than four-year college enrollment rate.⁴¹

³⁹Japan’s current fertility rate has been around 1.3, far below the population replacement level. In addition, the old-age dependency ratio is more than 50% and is projected to reach 80% in 2050, which is by far the highest among the OECD countries (See, <https://data.oecd.org/>).

⁴⁰See, https://www.mext.go.jp/a_menu/koutou/hutankeigen/detail/__icsFiles/afieddfile/2019/05/17/1417025_02_1.pdf (available only in Japanese).

⁴¹According to the Basic School Survey by the MEXT, the enrollment rate for some college was 4% in 2022 and is declining over the past thirty years.

To see this, I use the cross-country data provided by the OECD⁴² and define the *subsidization rate* for a specific education category e as follows:

$$s_e = \frac{y_e^{pub}}{y_e^{pri} + y_e^{pub}},$$

where y_e^{pri} and y_e^{pub} represent the private and public spending on education e , both represented as a share of the GDP. Public spending includes expenditures on educational institutions and educational-related subsidies for households or students. Private spending refers to expenditures financed by households and other private entities. Private spending includes expenditures on school but excludes those outside educational institutions (e.g., textbooks purchased by households, private tutoring, and student living costs).⁴³ In other words, the subsidization rate indicates what fraction of (potential) school-related costs are funded by the government.

As Table A.1 shows, Japan subsidizes more than 90% of the costs for primary-to-secondary education, which is higher than the OECD average. On the contrary, the subsidization rate for tertiary education was only 32%, which is the second lowest among OECD countries and less than half of the OECD average of 70%. This fact shows plenty of room for increasing subsidies for college students, which also has driven recent policy discussions on introducing and expanding education subsidies for college students.

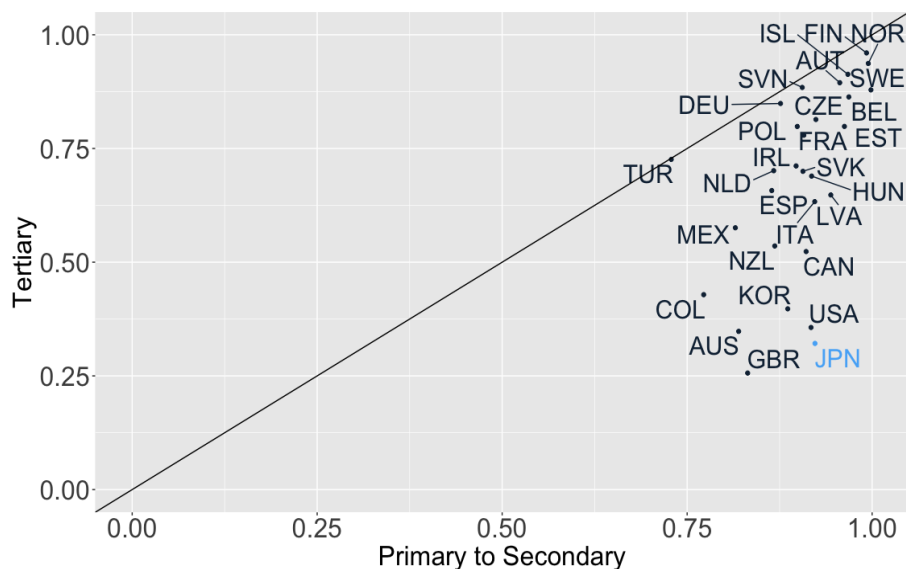


Fig A.1: Subsidization rate for each education category (OECD, 2018).

Fig A.2 plots the ratio of per-student government expenditures on tertiary education to per-capita GDP. On average, the government expenditures for each student enrolling in tertiary education amount to 33% of per-capita GDP, but the Japanese government only does to 21%.

⁴²I use 2018's data given that it is the latest year in which data on a significant number of countries is available.

⁴³For more details, see the OECD data (<https://data.oecd.org/>).

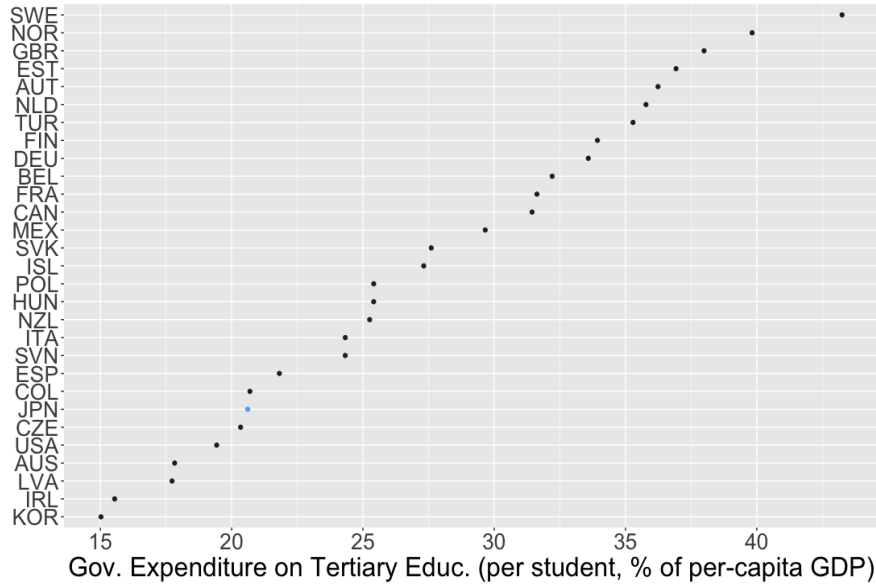


Fig A.2: Per-student government spending on tertiary education to per-capita GDP (OECD, 2016-17).

Fact 2. *Average parental transfers to college students amount to 65% of the average total education expenditures up to high school graduation.*

I use two data sets: (1) the Survey of Children’s Learning Expenses (SCLE, 2021) and (2) the Student Life Survey (SLS, 2018), both cross-sectional household surveys and conducted by the Ministry of Education, Culture, Sports, Science and Technology (MEXT).⁴⁴ The SCLE covers more than 53,000 students from preschool to high school and reports the per-student average education expenditure for each expenditure category and education stage (i.e., preschool, elementary, junior high, and high school). The expenditure category includes not only school-related ones (e.g., tuition fees and textbooks) but also extracurricular activities (e.g., cram school, music, arts, and sports). In addition to that, I use the SLS for parents’ expenditures when their children enroll in college.⁴⁵

⁴⁴See, https://www.mext.go.jp/b_menu/toukei/chousa03/gakushuuhi/1268091.htm for the SCLE and https://www.jasso.go.jp/statistics/gakusei_chosa/__icsFiles/afieldfile/2021/03/09/data18_all.pdf (in Japanese) for the SLS.

⁴⁵In 2018, the SLS correct answers from 43,394 students attending tertiary education, including college, some college, and graduate school.

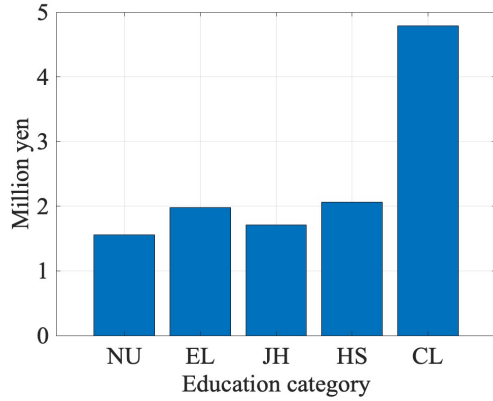


Fig A.3: Average education expenditures.

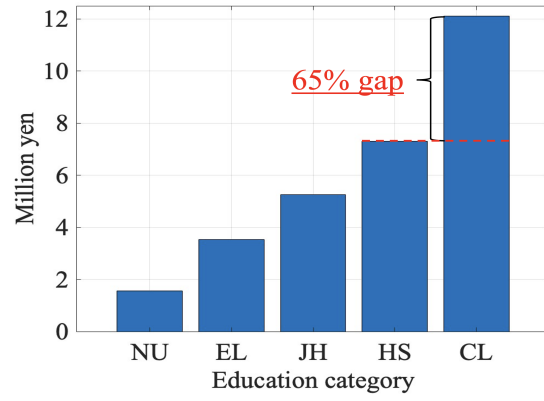


Fig A.4: Cumulative education-expenditures.

Fig A.3 shows the average per-child expenditure for each education category until completion. “NU,” “EL,” “JH,” “HS,” and “CL” each stand for nursery school or preschool, elementary school, junior high school, high school, and college, only consisting of a four-year college. All expenditures are conditional on enrollment in the education stage. Fig A.4 represents the cumulative education expenditures, showing that the expenditure jumps up when children attend college. Given that the high school graduation rate is approximately 100% in Japan, the figure tells us that raising one child on average takes the education costs of 7.31 million yen, which amounts to 4.5% of the individuals’ average lifetime labor earnings.⁴⁶ If their children attend college, they have to spend another 4.8 million yen, meaning there is about a 65% increase in education costs if parents send their children to college. This expensiveness of college education can, at least partly, be attributed to the fact that Japan is one of the least in subsidizing tertiary education while subsidizing a sizable portion of schooling costs up to secondary education. These observations establish Fact 2: A significant financial cost gap exists between those who have children enrolled in college and those who do not.

Fact 3. *About 80% of parents desire a college education for their children.*

We return to the NFS (2015), which asked respondents about the desired education level for their children. Fig A.5 summarizes the results by wife’s age. Here, “SC” stands for some college. The figure shows that approximately 75% of the parents desire a college education for their children, which is significantly higher than the college enrollment rate, which was 56.6% in 2022. This observation suggests that, although college enrollment in Japan has been far below 100%, education costs for a college education can be relevant not only for a specific part of the population because many parents would like to send their children to college.

⁴⁶I construct a proxy of the average individual’s lifetime earnings based on the 2022 Basic Survey on Wage Structure (BSWS) by the Ministry of Health, Labour, and Welfare (MHLW). First, I compute the average monthly earnings of ordinary workers for each age unconditional on any other characteristics such as sex and education. Then, I sum up the average earnings for each age, which amounts to about 160 million yen, and regard it as a proxy of the individuals’ average lifetime labor earnings.

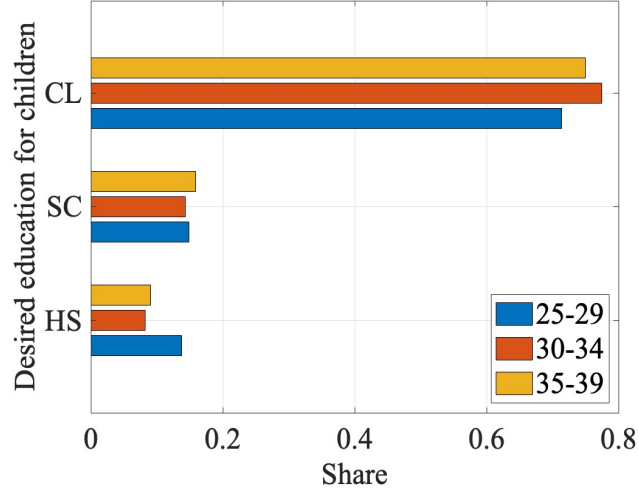


Fig A.5: Wives' intention for children's education attainment.

Fact 4. *Couples stop at fewer children than their ideal number primarily due to financial costs.*

This fact is drawn from the National Fertility Survey (NFS), which is provided by the National Institute of Population and Social Security Research (IPSS).⁴⁷ The NFS is a cross-sectional household survey that asks respondents about their preferences or intentions regarding fertility, marriage, child-raising, and education, as well as their basic information, including their education, age, and income. It is conducted nearly every five years, and the latest survey available is in 2015, which collected 5,334 couples in which the wife is aged 18 to 49. Hereafter, I present the results focusing on married couples in which the wife is aged 25 to 39 years, leaving 2,420 couples.⁴⁸ According to the NFS, a non-negligible gap exists between the ideal and planned numbers of children. In the 2015 survey, the ideal number of children for wives aged 25 to 39 was on average 2.38, whereas the planned number was 2.16. Fig A.6 represents the distribution of the ideal and planned numbers of children, where blue (red) bars indicate the share of wives who desire (plan) to have each number of children from zero to more than five. This figure suggests that the gap originates from the downward revision of the ideal at the intensive margin. There is no significant share gap between those whose ideal number is zero and those whose planned number is zero, and a substantial fraction of wives who desire three children end up with one or two children.

⁴⁷See, https://www.ipss.go.jp/site-ad/index_english/survey-e.asp.

⁴⁸Its 53.1% of the sample consists of couples in which the wife is over age 40, so the sample size shrinks if we target the younger couples. I focus on wives under age 39 because here I am interested in the fertility intention of those in the stage of fertility decision. Note that the cohort fertility rate is stable after age 40 for any cohort in Japan. See, for example, p7 of <https://www.mhlw.go.jp/toukei/saikin/hw/jinkou/tokusyu/syussyo07/dl/gaikyou.pdf> (in Japanese). Excluding those aged 18 to 24 does not affect the result significantly because they consist only of 1.5% of the observations for wives aged 18 to 49.

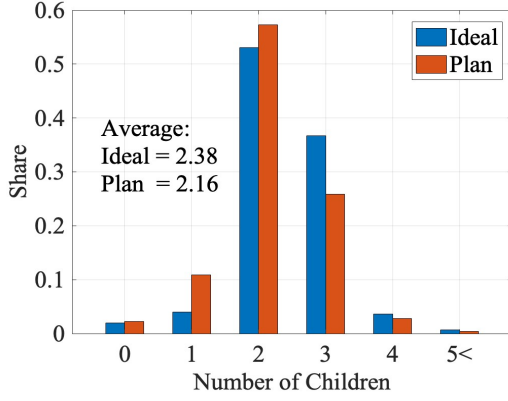


Fig A.6: Distribution of ideal and planned numbers of children.

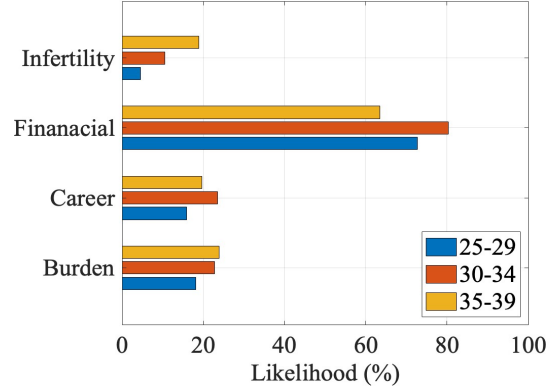


Fig A.7: Reasons for the gap between the ideal and planned number of children.

Why do people abandon their ideal number of children? The NFS asks them to pick their reasons among several options, and the result indicates that they are most likely to choose the financial reason: “raising children and education are too expensive.” Fig A.7 represents the likelihood⁴⁹ that wives of each age group abandon their ideal number of children for a particular reason, and “Financial” represents the financial reason. Aside from this, “Career” represents “it would interfere with my job,” and “Burden” represents “I would not handle the psychological and physical burden” (arising from achieving the ideal number). On average, more than 75% of them chose the financial reason, which dominates other reasons, such as “Career” and “Burden,” chosen by only 20% of them.

We observe the same even using the JPSC, which asks respondents about their preferences or intentions regarding fertility, such as “Do you want more children in the future?” If they answer “No,” it then asks the reasons from fourteen options, which they can choose multiple. The main options are: (1) having and educating children takes too much financial cost; (2) I would rather spend more time with my husband or for myself; (3) I want to continue my job; and (4) I cannot expect my husband’s effort in child-rearing. Fig A.8 reports the likelihood for each of the four reasons, where I refer to (1), (2), (3), and (4) as *Financial*, *Time*, *Job*, and *Husband*, respectively. This figure indicates that more than 40% of parents cease reproduction due to the financial reason, which is the most important among other reasons.

⁴⁹I use the term “likelihood” instead of “share” because it allows respondents to choose multiple options.

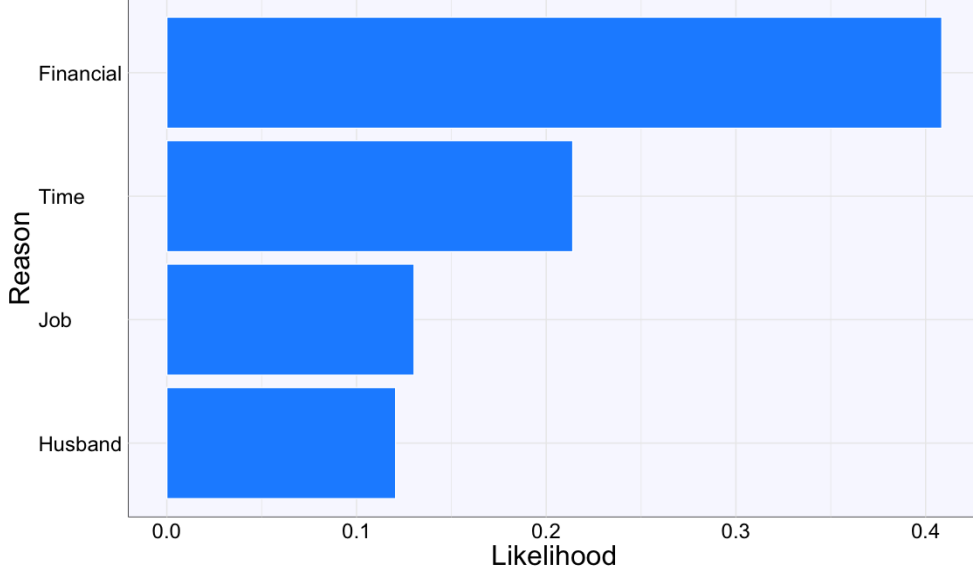


Fig A.8: Likelihood for reasons why parents cease reproduction. *Note:* Each of the four reasons, *Financial*, *Time*, *Job*, and *Husband*, represent: (1) having and educating children takes too much financial cost; (2) I would rather spend more time with my husband or for myself; (3) I want to continue my job; and (4) I cannot expect my husband’s effort in child-rearing.

To sum up, there is a non-negligible gap between the ideal and planned numbers of children for couples, and a substantial fraction of them answer the reason as “raising children and education are too expensive.” These results establish the first fact: couples are most likely to abandon having an ideal number of children because of financial costs.

Fact 5. *Children’s college enrollment is associated with a higher likelihood of parents stopping at fewer children due to the financial reason.*

This fact links the above four facts and suggests that college education costs are an important determinant of fertility choices in Japan. To see this, I conduct the following regression to examine the relationship between fertility *intentions* and children’s college enrollment:

$$\Pr(y_i = 1) = L(\alpha + \beta ChildCL_i + \mathbf{x}_i\boldsymbol{\theta} + \varepsilon_i),$$

Here, y_i is a dummy variable taking one if a couple i decides to stop having an additional child due to the financial reason. The results are summarized in Table 8.

	(1)	(2)
$ChildCL_i$	0.515*** (0.186)	0.870*** (0.222)
Personal characteristics	No	Yes
Observations	692	

Table 8: Children’s college enrollment and fertility intentions of women aged 24-39. *Note:* Values in parentheses represent standard errors. *** denotes statistical significance at the 1 percent level.

The first column presents the estimates where I control only for the children's college enrollment indicator. In the second column, I add couples' characteristics, such as educational backgrounds. The first and second columns report that the log odds ratio for couples choosing the financial reason to stop having children increases by 0.515 and 0.870 points for couples whose children will enroll in college. Both are statistically significant, and the latter implies a 2.4 times higher odds ratio for stopping having an additional child due to financial reasons.

B Analytical Models

B.(i) Fertility Differentials

B.(i).(A) The Price of Time Theory

Following [Jones, Schoonbroodt, and Tertilt \(2010\)](#), I consider the following problem:

$$\begin{aligned} \max_{c,n} \left\{ \frac{c^{1-\sigma}}{1-\sigma} + \theta \frac{n^{1-\sigma}}{1-\sigma} \right\} \\ \text{s.t.} \\ c = w \cdot (1 - n \cdot \chi) \end{aligned}$$

Here, c , n , w , and χ denote consumption, fertility, an exogenous wage, and time costs of children. The optimal fertility decision n^* is given as:

$$n^* = \frac{1}{w^{\frac{1-\sigma}{\sigma}} \cdot (\chi/\theta)^{\frac{1}{\sigma}} + \chi},$$

meaning that the negative income-fertility relationship holds if $\sigma > 1$ (i.e., the substitutability between consumption and children are high enough). Because having children is time-consuming, its (opportunity) cost is higher for high-income households. At the same time, high-income households, by definition, have higher incomes and afford to have more children. If the substitutability between consumption and children is sufficiently high, the substitution effect dominates the income effect, implying the negative income-fertility relationship.

B.(i).(B) The Quantity-Quality Trade-off

Following [De La Croix and Doepke \(2003\)](#), I consider the following problem:

$$\begin{aligned} \max_{c,n,e} \{ \ln(c) + \theta \ln(nq) \} \\ \text{s.t.} \\ c + b \cdot n \cdot e = w \cdot (1 - n \cdot \chi), \\ q = (\eta + e)^\gamma, \\ c, n > 0, e \geq 0. \end{aligned}$$

Here, the choice variables c , n , and e represent the consumption, fertility, and per-child education investment. The investment e is transformed into the “quality” of children, q , according to a innate ability production function $q = (\eta + e)^\gamma$. The parameter $\gamma \in (0, 1)$ governs the marginal gains of investment, and $\eta > 0$ implies that the quality of children takes a positive value even if parents make no investments. Parameters χ and b represent the time cost of having a child and a unit (monetary) cost of investment.

The optimal solutions for the fertility and education investment are given as

$$e^* = \frac{\gamma w \chi / b - \eta}{1 - \gamma}, \quad (7)$$

$$n^* = \frac{1}{1 + \theta} \cdot \frac{1 - \gamma}{\chi - \eta b / w}. \quad (8)$$

There are two critical observations in the optimal choices (7) and (8): (1) the optimal number of children is decreasing in wage level w , and (2) the optimal number of children is decreasing in the investment efficiency γ whereas the optimal investment is increasing in γ . These two observations capture the main theoretical explanations for the negative relationship between income and fertility: (1) having children is more costly for higher-income households, provided that having children is time-intensive, and (2) higher-income households have higher demands for child quality or an advantage in parental investments.⁵⁰

My full quantitative model captures these two channels, contributing to replicating the fertility differentials between college and high school graduates observed in the data. First, the opportunity costs of having children are more significant for college graduates, making child-bearing more costly for them. This is because having a child requires parents to incur a fixed amount of time, and college graduates’ wage rates are higher than those of high school graduates. Second, college graduate parents are more willing to pay for the IVT, making a child more costly. This is because the children’s innate ability and psychic costs of education, (h_k, ϕ_k) , govern education returns and the willingness to attend college, which in turn governs the marginal gains from IVT for parents. In particular, the psychic costs of children correlate with those of parents, so college graduate parents derive higher utility from the IVTs than high school graduate parents.

B.(ii) More on Illustrative Model in Section 3

B.(ii).(A) Proof of Proposition 1

I first take the first-order condition with respect to n as follows:

$$\begin{aligned} \underline{a}^{-1} \cdot b'(n) &= \mathbb{E}V'(c) \\ &= \frac{1}{2}[V'(\bar{I} + \nu - n \cdot \underline{a}) + V'(\bar{I} - \nu - n \cdot \underline{a})] \end{aligned} \quad (9)$$

To capture the relationship between n and ν , I then express n as a function of ν and take the first-order derivatives of the both sides of equation (9) with respect to ν as

⁵⁰For more details, see [Jones et al. \(2010\)](#).

follows:

$$\begin{aligned}
2\underline{a}^{-1}b''(n(\nu))\frac{dn}{d\nu} &= V''(\bar{I} + \nu - \underline{a} \cdot n(\nu)) - V''(\bar{I} - \nu - \underline{a} \cdot n(\nu)) \\
&\quad - \underline{a} \cdot \frac{dn}{d\nu} [V''(\bar{I} + \nu - \underline{a} \cdot n(\nu)) + V''(\bar{I} - \nu - \underline{a} \cdot n(\nu))] \\
\Leftrightarrow 2\underline{a}^{-1}b''(n(\nu))\frac{dn}{d\nu} &+ \underline{a} \cdot \frac{dn}{d\nu} [V''(\bar{I} + \nu - \underline{a} \cdot n(\nu)) + V''(\bar{I} - \nu - \underline{a} \cdot n(\nu))] \\
&= V''(\bar{I} + \nu - \underline{a} \cdot n(\nu)) - V''(\bar{I} - \nu - \underline{a} \cdot n(\nu)).
\end{aligned}$$

Finally, we have the stated result as follows:

$$\frac{dn}{d\nu} = \frac{\overbrace{V''(\bar{I} + \nu - \underline{a} \cdot n(\nu)) - V''(\bar{I} - \nu - \underline{a} \cdot n(\nu))}^{>0 \text{ } (\because V''' > 0)}}{\underbrace{2\underline{a}^{-1}b''(n(\nu)) + \underline{a} \cdot [V''(\bar{I} + \nu - \underline{a} \cdot n(\nu)) + V''(\bar{I} - \nu - \underline{a} \cdot n(\nu))]}_{<0 \text{ } (\because b'' < 0 \text{ and } V'' < 0)}} < 0$$

B.(ii).(B) Proof of Proposition 2

Recall that the optimal fertility $n(\varepsilon)$, given the policy parameter ε , is characterized by

$$\begin{aligned}
n(\varepsilon) &= \{\mathbb{E}[a(I; \varepsilon) \cdot V'(c(\varepsilon))]\}^{-\frac{1}{\sigma}} \\
&= \{\mathbb{E}a(I; \varepsilon) \cdot \mathbb{E}V'(c(\varepsilon)) - \text{cov}(\mathbb{E}a(I; \varepsilon), \mathbb{E}V'(c(\varepsilon)))\}^{-\frac{1}{\sigma}}
\end{aligned}$$

where $a(I; \varepsilon) = \underline{a} - \bar{g}(I; \varepsilon)$. Taking the first derivative of fertility with respect to ε directly is not straightforward. Instead, I show that non-positive derivative ($n'(0) \leq 0$) leads to a contradiction with the optimality conditions, which in turn implies that $n'(0) > 0$.

To begin, I take the ratio of $n(\varepsilon)$ and $n(0)$, which is given by:

$$\frac{n(\varepsilon)}{n(0)} = \left\{ \frac{\mathbb{E}V'(c(\varepsilon))}{\mathbb{E}V'(c(0))} - \underbrace{\text{cov}(\bar{g}(I; \varepsilon), V'(c(\varepsilon)))}_{>0} \cdot B \right\}^{-\frac{1}{\sigma}}, \quad (10)$$

where $B = (\underline{a} \cdot \mathbb{E}V'(c(0)))^{-1} > 0$ since $V' > 0$.

First, $\text{cov}(\bar{g}(I; \varepsilon), V'(c(\varepsilon)))$ is positive because, at higher (lower) income levels, $\bar{g}(I; \varepsilon) = (\mathbb{E}I - I) \cdot \varepsilon$ is lower (higher), while $V'(c(\varepsilon))$ is smaller (larger) due to the assumption that $V'' < 0$.

Next, I take the first-order partial derivative of $\mathbb{E}V'(c(\varepsilon))$ with respect to ε , evaluated at $\varepsilon = 0$ (i.e., an economy without subsidy), which is given by:

$$\begin{aligned}
\left. \frac{\partial \mathbb{E}V'(c(\varepsilon))}{\partial \varepsilon} \right|_{\varepsilon=0} &= \mathbb{E}\{V''(c(0)) \cdot ((\mathbb{E}I - I) \cdot n(0) - a(I; 0) \cdot n'(0))\} \\
&= \underbrace{\mathbb{E}V''(c(0))}_{<0} \cdot \underbrace{\mathbb{E}(\mathbb{E}I - I) \cdot n(0)}_{=0} + \underbrace{\text{cov}(V''(c(0)), (\mathbb{E}I - I) \cdot n(0))}_{<0} \\
&\quad - \underline{a} \cdot n'(0) \cdot \mathbb{E}V''(c(0)) \\
&= \underbrace{\text{cov}(V''(c(0)), (\mathbb{E}I - I) \cdot n(0))}_{<0} - \underline{a} \cdot n'(0) \cdot \underbrace{\mathbb{E}V''(c(0))}_{<0},
\end{aligned} \tag{11}$$

where the sign of $\text{cov}(V''(c(0)), (\mathbb{E}I - I) \cdot n(0))$ is negative because, with realization of higher (lower) income levels, V'' takes higher (lower) values because of the property $V''' > 0$, whereas $(\mathbb{E}I - I) \cdot n(0)$ takes lower (higher) values, and vice versa.

Note that the sign of the term on the right-hand side of equation (11), $n'(0)$, is of our interest. Suppose that $n'(0)$ were zero or negative, meaning that introducing an income-tested subsidy leads to a lower fertility. Then, the equation (11) implies that $\mathbb{E}V'(c(\varepsilon)) < \mathbb{E}V'(c(0))$ for some small $\varepsilon \in \mathcal{E}$ and that

$$\frac{\mathbb{E}V'(c(\varepsilon))}{\mathbb{E}V'(c(0))} < 1. \tag{12}$$

Equations (10) and (12) imply that

$$\frac{n(\varepsilon)}{n(0)} > 1 \Leftrightarrow n(\varepsilon) > n(0),$$

for some small ε . This contradicts $n'(0) \leq 0$, and hence such a case cannot occur. Therefore, we obtain

$$n'(0) > 0,$$

as stated.

B.(ii).(C) Decomposing Direct Effects

Instead of the subsidy function $\bar{g}(I; \varepsilon) = (\mathbb{E}I - I) \cdot \varepsilon$ in the illustrative model, I consider a new function $\bar{g}(I; \psi, \tau) = \psi/I + \tau$, where ψ and τ are new policy parameters. Let $n(\psi, \tau)$ be optimal fertility choices given policy parameters (ψ, τ) . Then, the effect of income-tested subsidy scheme, $\bar{g}(I, \psi, 0)$, compared with an economy without subsidy, $\bar{g}(I, 0, 0)$, can be expressed as follows:

$$\log \left(\frac{n(\psi, 0)}{n(0, 0)} \right) = \underbrace{\log \left(\frac{n(0, \tau)}{n(0, 0)} \right)}_{=\text{price effect}} + \underbrace{\log \left(\frac{n(\psi, 0)}{n(0, \tau)} \right)}_{=\text{insurance effect}}, \tag{13}$$

where a parameter τ satisfies a condition $\mathbb{E}\bar{g}(I, \psi, 0) = \bar{g}(I, 0, \tau)$ for a given $\psi \geq 0$. This condition means that the expected value of the subsidy is neutral between the income-tested subsidy scheme $\bar{g}(I, \psi, 0)$ and the uniform subsidy scheme $\bar{g}(I, 0, \tau)$. The price effect captures the effects of reducing the expected monetary costs (or “price”) of children, while

the insurance effect captures the effects of smoothing marginal utilities as discussed in the illustrative model, holding the expected value of the subsidy constant.

I use (13) to decompose the direct effect observed in the full quantitative model. In practice, I construct a grant function that provides the expected value of the cash transfers to each household, given the probability of the household being eligible for the subsidy. I then compute the effects of such uniform subsidy on fertility, which is the price effect. The insurance effect is then computed as the difference between the overall direct effect and the price effect.

C Quantitative Model

C.(i) Value Functions

Early Working Stage: The remaining component of the value functions during the education stage, V^w , represents value functions in the early working stage (before fertility stage) for households aged $j \in \{J_E, \dots, J_F - 1\}$. The state variables for this stage consists of asset a , age j , idiosyncratic component of labor productivity z , education level e , and innate ability h . The uncertainty in this stage is only about the next period's productivity, which is denoted by z' following a Markov process $\pi_z(z', z)$. Households choose consumption, leisure, and savings given the state variables. The value function is formulated as follows:

$$\begin{aligned}
V^w(a, j, z; e, h) = & \max_{c, l, a'} \{ u(c, l) \\
& + \begin{cases} \beta \mathbb{E}_{z'} [V^f(a', z', e, h)] & \text{if } j = J_F - 1 \\ \beta \mathbb{E}_{z'} [V^w(a', j + 1, z'; e, h)] & \text{otherwise} \end{cases} \} \\
\text{s.t.} & \\
(1 + \tau_c)c + a' = & (1 - \tau_w)w_e \eta_{j, z, e, h} (1 - l) + \psi + (1 + (1 - \tau_a)r)a, \\
z' \sim \pi(z', z), & a' \geq -\underline{A}.
\end{aligned} \tag{14}$$

Value functions V^f and V^r represent those for the fertility stage and retirement stage, which are formulated in Section 4.3.

Late Working Stage: At this stage for households aged $j \in \{J_{IVT}, \dots, J_{R-1}\}$, the value function is formulated as:

$$\begin{aligned}
V^w(a, j, z; e, h) &= \max_{c, l, a'} \{ u(c, l) \\
&\quad + \begin{cases} \beta[V^r(a', j+1)] & \text{if } j = J_R - 1 \\ \beta \mathbb{E}_{z'}[V^w(a', j+1, z'; e, h)] & \text{otherwise} \end{cases} \} \\
&\text{s.t.} \\
(1 + \tau_c)c + a' &= (1 - \tau_w)w_e \eta_{j, z, e, h}(1 - l) + \psi + (1 + (1 - \tau_a)r)a, \\
z' &\sim \pi(z', z), \\
a' &\geq \begin{cases} 0 & \text{if } j = J_R - 1, \\ -\underline{A} & \text{otherwise.} \end{cases}
\end{aligned} \tag{15}$$

Here, V^r denotes the value function for the retirement stage, as described below.

Retirement Stage: At the beginning of age J_R , households are forced to retire from the labor market. After that, they spend all their time on leisure and make consumption-saving decisions. Two additional points differ from previous choice problems: they receive the pension benefit p from the government and face uncertainty about the next period's survival. The value function for the retirement stage is formulated as follows:

$$\begin{aligned}
V^r(a, j) &= \max_{c, a'} u(c, 1) + \beta \xi_{j, j+1} V^r(a', j+1) \\
&\text{s.t.} \\
(1 + \tau_c)c + a' &= p + (1 + (1 - \tau_a)r)a + \psi, \\
a' &\geq 0.
\end{aligned}$$

C.(ii) Equilibrium Definition

Let $\mathbf{x}_j^e$ be an age-specific state vector for agents with education level $e \in \{HS, CL\}$ and $\mu_j^e(\mathbf{x}_j^e)$ be the measure of agents with state vector $\mathbf{x}_j^e$. Let $I_l(h, I)$ and $I_g(h, I)$ be indicator functions for loans and grants, respectively, returning 1 if students with (h, I) are eligible and 0 otherwise.

Given exogenous parameters and policy rules $\{\tau_a, \tau_c, \iota, \iota_s, B, S, \psi, p, I_l(h, I), I_g(h, I)\}$, a *stationary recursive competitive equilibrium* consists of

- value functions $\{V_{g0}, V_{g1}, V_{g2}, V^w, V^f, V^{wf}, V^{IVT}, V^r\}$,
- policy functions for consumption, savings, leisure $\{c_j^e(\mathbf{x}_j^e), a_j^e(\mathbf{x}_j^e), l_j^e(\mathbf{x}_j^e)\}_{j=J_E}^J$, working hours $\{h_j^e(\mathbf{x}_j^e)\}_{j=J_E}^{J_R}$, fertility $\{n_{J_F}^e(\mathbf{x}_{J_F}^e)\}$, IVT $\{a_{J_{IVT}}^e(\mathbf{x}_{J_{IVT}}^e)\}$, and college enrollment $\{e_{J_E}(\mathbf{x}_{J_E})\}$,
- prices (r, w_{HS}, w_{CL}) ,
- labor income tax rate τ_w ,
- aggregate quantities (K, L_{HS}, L_{CL}) ,
- measures for households $\{\mu_j^e(\mathbf{x}_j^e)\}_{j=J_E}^J$,

such that:

1. The decision rules of students, workers, and retired households solve their problems, and $\{V_{g0}, V_{g1}, V_{g2}, V^w, V^f, V^{wf}, V^{IVT}, V^r\}$ are the associated value functions.
2. The representative firm maximizes its profit and optimally chooses capital and labor inputs:

$$r + \delta = \alpha \cdot Z \cdot \left(\frac{K}{L}\right)^{\alpha-1}, \quad (16)$$

$$w_{HS} = \tilde{Z} \cdot \omega_{HS} \cdot L_{HS}^{\chi-1}, \quad (17)$$

$$w_{CL} = \tilde{Z} \cdot \omega_{CL} \cdot L_{CL}^{\chi-1}, \quad (18)$$

where

$$L = [\omega_{HS} \cdot (L_{HS})^\chi + \omega_{CL} \cdot (L_{CL})^\chi]^{1/\chi},$$

$$\tilde{Z} = (1 - \alpha) \cdot Z \cdot \left(\frac{K}{L}\right)^\alpha \cdot [\omega_{HS} \cdot (L_{HS})^\chi + \omega_{CL} \cdot (L_{CL})^\chi]^{1/\chi-1}.$$

3. The labor market for each skill $e \in \{HS, CL\}$ clears:

$$L_e = \sum_{j=J_E}^{J_R} \int_{\mathbf{x}_j^e} \eta_j^e(\mathbf{x}_j^e) \cdot h_j^e(\mathbf{x}_j^e) d\mu_j^e(\mathbf{x}_j^e), \quad (19)$$

where $\eta_j^e(\mathbf{x}_j^e)$ represents the labor efficiency of agents with a state vector $\mathbf{x}_j^e$.

4. The capital market clears:

$$K = \sum_{j=J_E}^J \sum_e \int_{\mathbf{x}_j^e} a_j^e(\mathbf{x}_j^e) d\mu_j^e(\mathbf{x}_j^e). \quad (20)$$

5. The government budget is balanced:

$$\tau_c \cdot C + \tau_w \cdot (L_{HS} + L_{CL}) + \tau_a \cdot K + Q = p \cdot \mu_{old} + (\iota - \iota_s) \cdot K_s + G + \psi + B \cdot \mu_{j \leq 17} + S,$$

where

$$\begin{aligned}
C &= \sum_{j=J_E}^J \sum_e \int_{\mathbf{x}_j^e} c_j(\mathbf{x}_j^e) d\mu_j^e(\mathbf{x}_j^e), \\
Q &= \sum_{j=J_R+1}^J \frac{1 - \zeta_{j-1,j}}{\zeta_{j-1,j}} \sum_e \int_{\mathbf{x}_j^e} a_j^e(\mathbf{x}_j^e) d\mu_j(\mathbf{x}_j^e), \\
p \cdot \mu_{old} &= \sum_{j=J_R}^J \sum_e \int_{\mathbf{x}_j^e} p d\mu_j^e(\mathbf{x}_j^e), \\
K_s &= \int_{\mathbf{x}^s} \max\{0, -a^s(\mathbf{x}^e)\} \cdot I_l(h, I) d\mathbf{x}^s \\
G &= \int_{\mathbf{x}^s} g(h, I) \cdot I_g(h, I) d\mathbf{x}^s \\
B \cdot \mu_{j \leq 17} &= \sum_{j=J_F}^{J_{IVT}-1} \sum_e \int_{\mathbf{x}_j^e} B \cdot n d\mu_j^e(\mathbf{x}_j^e),
\end{aligned}$$

where $\mathbf{x}^s$, $\mu^s(\mathbf{x}^s)$, and $\{a^s(\mathbf{x}^s)\}$ represent a state vector for college students, measure of college students, and students' policy function for saving, respectively.

6. Distributions (measures) and households' behavior are consistent.

C.(iii) Discussion of Model Assumptions

C.(iii).(A) Pre-College Human Capital Formation

In this model, parental expenditures made before their children graduate from high school (q) do not contribute to human capital formation and therefore do not affect the children's initial endowment. While a recent study by [Krueger et al. \(2024\)](#) emphasizes the importance of modeling endogenous pre-college human capital formation when evaluating education policies, I argue that this omission is not a major concern in the context of Japan.

To see why, note that the key insight of [Krueger et al. \(2024\)](#) is the complementarity between college- and pre-college-subsidies, accounting for the possibility of college dropouts: college subsidies improve college attendance rates, while pre-college subsidies enhance students' preparedness and increase the likelihood of successful college completion. However, given that the college dropout rate is quite low in Japan as explained in the main text, endogenizing pre-college human capital formation is likely of second-order importance when assessing the effects of college subsidies on completion rates. Abstracting from it therefore offers a gain in tractability without significant loss of realism.

That said, incorporating endogenous pre-college human capital formation remains an important avenue for future research, as it would allow for the analysis of a broader set of education policies and their implications for fertility decisions and macroeconomic outcomes. This paper takes a first step by studying the insurance effects of college subsidies on fertility and their macroeconomic implications, leaving the extension to endogenous pre-college human capital formation for future work.

C.(iii).(B) Fertility Timing

In this model, the timing of fertility choice is fixed at age J_F . While this assumption may arguably introduce an upward bias in the insurance role of the income-tested subsidy—since households may postpone childbearing under income uncertainty, and excluding this additional adjustment margin could overstate the effect—the direction of the bias is not clear-cut.

Indeed, fixing fertility timing could also lead to a downward bias, as income-tested subsidies may allow households to have children with less reliance on self-insurance. Without such subsidies, households might postpone childbirth until they are confident in their ability to self-insure—typically through accumulated savings. In contrast, when subsidies are available, the need for precautionary savings is reduced, which may encourage earlier childbearing. In this case, the insurance effect of the subsidy becomes more pronounced, as households bear a lower level of self-insurance than they would if they had postponed childbirth.

At the same time, the assumption may still lead to an upward bias for the same reason noted earlier: excluding the option to postpone childbirth may remove an important channel for mitigating the risk associated with having a child. However, reproductive ability declines exponentially with age, and delaying childbirth increases the risk of infertility, especially after age 35 (e.g., [Sommer, 2016](#)), which constitutes a cost of postponement. In my JPSC data sample, the average age at first birth is 30.01, suggesting that the marginal cost of postponement in terms of infertility risk is non-negligible. As a result, households have an incentive to avoid delaying childbirth, making income-tested subsidies a valuable form of insurance in such decisions. This cost of postponement is more pronounced for more educated parents, where college graduates giving birth later on average (31.34) than high school graduates (29.75). Since the insurance effect is more significant for college-educated parents, who are more likely to send their children to college, the potential upward bias introduced by fixing the timing of fertility is likely to be less severe than if the insurance effect were more concentrated among high school graduates.

D Computation and Calibration

D.(i) Computational Algorithm

D.(i).(A) Stationary Equilibrium

For any variable or distribution x , let $\tilde{x}$ and $\hat{x}$ represent its guessed and model-implied values. Also, let μ_j represent the distribution over state variables for age j . Dividing the computation process into three blocks makes it easier to understand. The first block is the outer loop, searching for equilibrium prices. The other two blocks are inner loops; one is the *optimization block*, solving household problems given prices, and another is the *distribution block*, searching for the stationary distributions given prices and policy functions obtained in the optimization block. The computational algorithm proceeds as follows:

1. Guess prices $\tilde{\mathbf{p}} = (\tilde{r}, \tilde{w}_{HS}, \tilde{w}_{CL})$.

2. *Optimization block:*

- Guess the value function for agents at the beginning of age $j = 18$ ($\tilde{V}_{g0}$).
- Given $\tilde{V}_{g0}$, solve backward from the period of IVT choice to that of the education choice, which gives the model-implied value function for V_{g0} , $\hat{V}_{g0}$.
- Check if

$$d(\tilde{V}_{g0}, \hat{V}_{g0}) < \varepsilon, \quad (21)$$

where $d(\cdot)$ and $\varepsilon > 0$ represent an arbitrary metric function and error tolerance. If (21) is not satisfied, update $\tilde{V}_{g0}$ and follow the same procedure until convergence. The correct V_{g0} pins down all value functions and policy functions given a set of prices $\tilde{\mathbf{p}}$.

3. *Distribution block:*

- Guess the distribution for age J_{IVT} . This $\tilde{\mu}_{J_{IVT}}$ and policy functions for IVT derive the implied distribution for agents aged 18, $\hat{\mu}_{18}$. Given $\hat{\mu}_{18}$ and policy functions, compute the implied distributions for age $j = 19, \dots, J_{IVT}$, and obtain $\hat{\mu}_{J_{IVT}}$.
- Check if

$$d(\tilde{\mu}_{J_{IVT}}, \hat{\mu}_{J_{IVT}}) < \varepsilon. \quad (22)$$

If (22) is not satisfied, update $\tilde{\mu}_{J_{IVT}}$ and follow the same procedure until convergence.

- After obtaining the correct distributions for age $j = 18, \dots, J_{IVT}$, compute the distribution for age $j_{IVT} + 1$ onward, using those distributions and policy functions.
4. After computing value functions, policy functions, and distributions, compute the implied quantities, $\hat{L}$ and $\hat{K}$ based on (19) and (20), which gives the implied prices $\hat{\mathbf{p}}$ based on $\hat{L}$, $\hat{K}$, (16), (17), and (18).
5. Check if

$$d(\tilde{\mathbf{p}}, \hat{\mathbf{p}}) < \varepsilon. \quad (23)$$

If (23) is not satisfied, update $\tilde{\mathbf{p}}$, return to the optimization block, and follow the same procedure until convergence.

D.(i).(B) Transition Dynamics

For any variable or distribution x , let $\tilde{x}$ and $\hat{x}$ represent its guessed and model-implied value (or distribution). Also, let $\mathbf{q}_t = (K_t, L_t)$ be a vector of aggregate quantities in a year t , and $\mathbf{q} = (\mathbf{q}_t)_{t=1, \dots, T}$ for some T . The computational algorithm for transition dynamics proceeds as follows:

1. *Preliminaries:* Set an arbitrarily large number for transition periods, T . Given aggregate quantities in the initial and final steady states, $\mathbf{q}_1$ and $\mathbf{q}_T$, guess a sequence of aggregate quantities, $\{\tilde{\mathbf{q}}_t\}_{t=1, \dots, T}$, where $\tilde{\mathbf{q}}_1 = \mathbf{q}_1$ and $\tilde{\mathbf{q}}_T = \mathbf{q}_T$, which pins down guesses for prices $\{\tilde{r}_t, \tilde{w}_t\}_{t=1, \dots, T}$. Also, guess a sequence of the labor income tax

rate $\{\tilde{\tau}_{l,t}\}_{t=1,\dots,T}$, where $\tilde{\tau}_{l,1}$ and $\tilde{\tau}_{l,T}$ are the tax rates at the initial and final steady states.

2. *Optimization*: Given $\{\tilde{r}_t, \tilde{w}_t\}_{t=1,\dots,T}$ and $\{\tilde{\tau}_{l,t}\}_{t=1,\dots,T}$, solve the optimization problem for each cohort born before period T .
3. *Distribution*: Given decision rules for each cohort obtained in the above optimization block, construct the implied distributions:
 - $\tilde{\mu}_a(j; t)$: (aggregate) savings distribution across age in year t .
 - $\tilde{\mu}_n(j; t)$: (aggregate) labor supply distribution across working-age in year t .
 - $\tilde{\mu}_{j|t}$: age distribution in year t .
4. Given the decision rules and the implied distributions, compute the model-implied aggregate quantities, $\hat{\mathbf{q}}$. Also, compute the implied government revenue and expenditures, denoted by $\hat{\mathbf{G}}_r = (\hat{\mathbf{G}}_{r,t})_{t=1,\dots,T}$ and $\hat{\mathbf{G}}_e = (\hat{\mathbf{G}}_{e,t})_{t=1,\dots,T}$.
5. Check if

$$d(\tilde{\mathbf{q}}, \hat{\mathbf{q}}) < \varepsilon$$

and

$$d(\hat{\mathbf{G}}_r, \hat{\mathbf{G}}_e) < \varepsilon.$$

If the first (second) condition is not satisfied, update $\tilde{\mathbf{q}}$ ($\{\tilde{\tau}_{l,t}\}_{t=1,\dots,T}$) and follow the same procedure until convergence.

D.(ii) More on Calibration and Validation

D.(ii).(A) Stochastic income process

I estimate the AR(1) process for college graduates and high school graduates separately, following the procedure of [Darulich and Kozłowski \(2020\)](#). First, I control for the age component for each education level, $\gamma_{j,e}$, and obtain the residual $u_{i,j,e}$ for an individual i aged j with education level e , which is considered the sum of two components as follows:

$$u_{i,j,e} = m_{i,j,e} + z_{i,j,e}.$$

The measurement error $m_{i,j,e} \sim N(0, \sigma_e^m)$ is normally distributed with mean 0 and variance σ_e^m , and $z_{i,j,e}$ is a persistent shock that follows an AR(1) structure:

$$\begin{aligned} z_{i,j,e} &= \rho_e^z z_{i,j-1,e} + \epsilon_{i,j,e} \\ \epsilon_{i,j,e} &\sim N(0, \sigma_e^z). \end{aligned}$$

The initial draw for z , $z_{i,0,e}$, follows $N(0, \sigma_e^{z_0})$. I estimate this process for each education group using a Minimum Distance Estimator, using the variance of income residuals, their first-order autocovariance, and the variance of their first differences as moments.

D.(ii).(B) Fertility–subsidy gradient

The regression equation for the fertility–subsidy gradient, discussed in Section 6.3, is given as follows:

$$\log(\text{Fertility}_{i,t}) = \alpha + \beta \text{Spending}_{i,t} + \mathbf{x}'_{i,t} \boldsymbol{\gamma} + \zeta_t + \eta_i + \varepsilon_{i,t},$$

where $\log(\text{Fertility}_{i,t})$ represents the log of the total fertility rate in country i in year t , $\text{Spending}_{i,t}$ is the government spending on tertiary education (per student / GDP per capita $\times 100$) in country i and year t , and $\mathbf{x}'_{i,t}$ includes GDP per capita, Gini coefficient, government expenditures on other education stages, and population share of working-age population. The estimation results are summarized in Table 9.

	(1)	(2)
$\text{Spending}_{i,t}$	$2.7e - 04^{***}$ ($7.5e - 05$)	$1.3e - 03^*$ ($7.6e - 04$)
Sample	Full	OECD
Observations	456	201

Table 9: Estimation results. *Note:* Values in parentheses represent standard errors. *** and * denote statistical significance at the 1 and 10 percent levels.

D.(ii).(C) Parameters

	High school	College
Age	+0.041	+0.048
$\text{Age}^2 \times 10,000$	−4.551	−5.364
ρ_e^z (persistence)	0.924	0.920
σ_e^z (volatility)	0.060	0.048

Table 10: Parameters governing the age profile and stochastic process for wages. *Note:* CL indicates the college graduate households where the husband or wife is a college graduate. HS represents the rest of the population.

Parameter	Value	Description	Source
<i>Taken from literature estimate</i>			
χ	0.70	Elasticity of substitution	Taniguchi and Yamada (2024)
τ_a	0.35	Capital income tax	Hansen and İmrohoroglu (2016)
τ_w	0.35	Labor income tax	Gunji and Miyazaki (2011)
<i>Taken directly from data</i>			
α	0.31	Capital share	JIP database
$\underline{A}$	3.60	Borrowing limit (/ Avg. income at 28)	FIES
$\underline{A}_s$	0.52	For students (/ Avg. income at 28)	SLS
p_{CL}	0.20	Tuition fees (/ Avg. income at 28)	JPSC
κ	0.044	Time costs	STULA
$\xi_{j,j+1}$	—	Survival prob.	Vital Statistics
p	0.31	Pension benefits (/ Avg. income at 28)	JPS
b	0.02	Cash transfers (/ Avg. income at 28)	MHLW
$\bar{t}$	0.21	Time in uni-related activities	SLS
τ_c	0.10	Consumption tax	
<i>Estimated directly from data</i>			
ρ_e^z	See Table 10	Persistence	JPSC
σ_e^z	See Table 10	Transitory	JPSC
$\{\gamma_{j,e}\}_{j,e}$	See Table 10	Wage growth in age	JPSC

Table 11: Parameters externally determined.

Parameter	Value	Moment	Data	Model
ν	1.0	Education sorting by ability		
γ	0.5	Curvature		
β	0.98	Discount factor		
A_s		public college spending relative to GDP	0.3%	0.3%
δ	0.07	Depreciation rate (backed out)		
μ	0.23	Work hours	0.33	0.30
$\bar{t}$	0.8	Income share of labor earnings	0.21	0.20
ι_s	0.054	Share of students using loans	0.44	0.34
ι	0.055	Household share with negative net worth	0.54	0.45
ω_{CL}	0.52	CL–HS wage ratio	1.36	1.48
ψ	0.01	Var(log disposable income)/Var(log gross income)	0.60	0.68
λ_q	0.62	Average transfer / Average income at age 28	0.07	0.07
λ_a	1.03	Average transfer / Average income at age 28	0.27	0.27
ω	1.71	Intergenerational mobility of education	See Table 4	
ρ_h	0.30	Intergenerational income elasticity	0.30	0.27
σ_h	0.65	Variance of log(income) at age 28	0.27	0.24
ψ_{CL}	20.8	College enrollment rate	0.377	0.376
α_{CL}	0.1	Log wage ratio (CL–HS) at age 28	0.34	0.38
β_{CL}	0.1	Var log wage for CL at age 28	0.14	0.24
b_1	0.49	Share of one child	0.16	0.15
b_2	0.53	Share of two children	0.55	0.61
b_3	0.55	Share of three children	0.22	0.24
b_4	0.56	Share of four or more children	0.02	0.00
Z	1.99	High school graduates’ wage (backed out)	1.0	1.0

Table 12: Parameters internally determined. *Note:* “HS” and “CL” indicate high school and college graduates.

E Counterfactual Experiments

E.(i) On the Welfare Analysis

The welfare analysis using models with endogenous fertility is not straightforward theoretically and philosophically. One of the well-known difficulties is that the standard concept of Pareto efficiency is not applicable to models with endogenous fertility (Goloso, Jones, and Tertilt, 2007).⁵¹ While recognizing these difficulties, this study discusses the welfare effects of the policy by computing the consumption equivalence under the veil of ignorance under the benchmark economy relative to the new steady state,⁵² which is a useful

⁵¹This is because models with endogenous fertility require a welfare comparison between two different sets of individuals. Goloso, Jones, and Tertilt (2007) then propose two efficiency concepts that can apply to models with endogenous fertility. For example, the $\mathcal{A}$ –efficiency proposed by them focuses on individuals being alive in both economies. This concept is particularly beneficial if we examine the optimal policy as they show that the unique solution for the planning problem considering the utility of the existing agents (i.e., those who are alive just before the policy is introduced) corresponds to the $\mathcal{A}$ –efficiency.

⁵²The related work Zhou (2022) adopts the same welfare criteria in its steady-state analysis. In its transition analysis, Zhou (2022) also computes the average welfare of the existing households already alive before the policy changes to evaluate the $\mathcal{A}$ –efficiency of the policy.

standpoint for considering the potential channel through which education subsidies affect economic welfare.

To formalize the measure, let $P \in \{0, 1, 2, \dots\}$ denote education subsidy schemes or policies with $P = 0$ representing the benchmark economy without grants. Next, let $V^P(\lambda)$ be the expected lifetime utility in stationary equilibrium for newborn agents with a consumption scaling parameter λ and policy P :

$$V^P(\lambda) = \int_{\mathbf{x}_1} V_{j=1}^P(\mathbf{x}_1; \lambda) d\mu(\mathbf{x}_1). \quad (24)$$

$\mu(\mathbf{x}_j)$ is the measure over the age-specific state space where $j \in \{1, \dots, J\}$ and $V_{j=1}^P(\mathbf{x}_1; \lambda)$ represents the expected utility for an agent aged $j = 1$ with a state vector $\mathbf{x}_1$:

$$V_{j=1}^P(\mathbf{x}_1; \lambda) = \mathbb{E} \left\{ \sum_{j=1}^J \beta^{j-1} \cdot u(c_j \cdot (1 + \lambda), l_j) + \sum_{j=J_F}^{J_{IVT}-1} \beta^{j-1} b(n) \cdot v(q_j) + \beta^{J_{IVT}-1} \lambda_a \cdot b(n) \cdot V_{g0}(\mathbf{x}_{J_{IVT}}) \right\}$$

Here, $\{c_j, l_j, q_j, n, a_{IVT}\}$ are optimal choices with the policy P and each state $\mathbf{x}_j$. Finally, the consumption equivalence with a policy P is given as a scalar λ satisfying the following equation:

$$V^0(\lambda) = V^P(0).$$

That is, the consumption equivalence λ makes the newborn agents indifferent between the new economy and the benchmark by scaling up the benchmark consumption by λ .

E.(ii) Decomposing the Long-run Effects

The long-run increases in fertility, college enrollment rates, output, and welfare can be broken down into direct effects and indirect effects. The direct effects capture the effects of introducing the policy holding macroeconomic objects—such as prices, tax rates, and distributions—fixed and can be further broken down into the insurance effect and price effect. The indirect effects capture the effects of the policy induced by changes in factor prices (i.e., GE effects), tax rates (i.e., Taxation effects), and household distribution (i.e., Distributional effects). To isolate each effect, I conduct a decomposition exercise as follows. I first solve an equilibrium by introducing the new grant function. Then, I solve household problems by replacing one of the four objects (grant function, prices, tax rate, and household distribution) in the benchmark with that in the new equilibrium. Note that this method does not guarantee that each implied effect adds up to the overall effects because all factors except grant function are endogenous and interconnected; however, my results indicate that each effect adds up roughly to the overall effects. I further decompose the direct effects into the insurance and price effects based on the method discussed in Section B.(ii).(C). Tables 6 and 14 summarize the decomposition results.

	Direct		Indirect			All
	Average	Insurance	GE	Tax	Dist.	
TFR (Δ %)	+0.8	+1.3	+1.0	0.0	-1.0	+3.1
HS (Δ %)	+0.1	+0.8	0.0	0.0	0.0	-0.3
CL (Δ %)	+2.4	+2.1	+2.4	0.0	0.0	+10.5

Table 13: Decomposition results. *Note:* Columns “Direct,” “GE,” “Tax,” and “Dist.” report the results when only grant function, prices, labor income tax rate, and distribution change, respectively. A column “All” indicates the results in the benchmark and the long-run equilibrium with the grants. Rows “CL share” represent the change in college enrollment rate in p.p. and rows “HS” and “CL” represent the percentage changes in fertility rates for high school and college graduates.

Fertility: For the long-run increase in the TFR, vital forces are direct and GE effects. If the grant is introduced while other variables are fixed as in the benchmark, the TFR increases by 2.1% (direct effects), which corresponds to two-thirds of the overall effects.

The GE effect also plays a role in accounting for the TFR increase. If the prices are set to the long-run equilibrium levels while other objects are fixed as in the benchmark, the college graduates’ fertility increases by 2.4%, and the TFR then increases by 1.0%. The key is the decline of the wage rate for college graduate workers, w_{CL} . In the long-run equilibrium with the grants, w_{CL} decreases by 2.2%. First, the greater supply of college graduate workers depress w_{CL} relative to w_{HS} . In addition, greater aggregate labor supply in efficiency units implies the lower marginal productivity of labor, decreasing w_{CL} . The lower w_{CL} implies the lower opportunity costs of having children for college graduates, making some have more children. Also, for college graduate parents, the lower w_{CL} reduces the (expected) utility of the IVTs, given that their children are likely to attend college, as discussed above. The lower utility of the IVTs also contributes to higher fertility because the marginal utility gains from investing in children’s “quality” (by making IVTs to send a child to college) are relatively lower than those from increasing its “quantity.”⁵³

Lastly, the distributional effect implies a 1.0% decline in the average fertility. The introduction of the grants increases the college enrollment rate, meaning that the share of college graduates increases in the long run. Given that college graduates have fewer children than high school graduates as discussed in Section 6.1, the greater share of college graduates means a greater share of those who tend to have fewer children, which can lead to a lower average fertility.⁵⁴ However, direct and GE effects are significant and lead to the higher TFR.

⁵³This fertility response of college graduates to the change in skill premium aligns with an empirical observation (Lehr, 2003).

⁵⁴This effect is comparable with a composition effect highlighted by Zhou (2022) in understanding the effects of public education subsidies on fertility. In his model, in the long run, public education subsidies increase the share of parents with higher human capital, who tend to have fewer children.

	Direct	GE	Tax	Dist.	All
CL share (Δ p.p.)	+2.5	-0.1	0.0	+2.2	+4.0
Output (Δ %)	-0.7	-0.1	-0.1	+1.4	+0.7
Welfare (%)	+2.6	-0.7	-0.2	+2.3	+5.1

Table 14: Decomposition results. *Note:* Columns “Direct,” “GE,” “Tax,” and “Dist.” report the results when only grant function, prices, labor income tax rate, and distribution change, respectively. A column “All” indicates the results in the benchmark and the long-run equilibrium with the grants. Welfare gains are represented in terms of consumption equivalence.

College Enrollment: For the long-run increase in the college enrollment rate, critical forces are direct and distributional effects. If the grant is introduced while other variables (i.e., prices, tax rate, and distribution) are fixed as in the benchmark, the college enrollment rate increases by 2.5 p.p., more than half of the overall increase. These direct effects capture the effects of relaxing financial constraints on college enrollment decisions. GE effects put downward pressure on college enrollment because the lower college premium reduces the incentive for college enrollment.

Note that these direct effects do not explain the overall effects, so we need other forces accounting for the increase in college enrollment rate; that turns out to be the distributional effects. The implied changes in the distribution in the long run, holding grant function, prices, and tax rate fixed as in the benchmark, lead to a 2.2 p.p. higher college enrollment rate. The most relevant factor is the change in the skill distribution of parents (i.e., the share of college graduates). The increase in the share of college graduates implies a higher share of those more likely to have children enrolling in college due to the intergenerational persistence of education. Then, the distributional change amplifies the short-run increase in college enrollment rate facilitated by the direct effects.

Output: The distributional effect is critical in accounting for the output increase. In the long run, the working-age population increases due to a higher equilibrium TFR, and the share of skilled workers increases. These forces increase the labor supply in efficiency units, increasing output in the long run.

In contrast, the direct and GE effects put downward pressure on output. As indicated in Table 14, the GE effects lower the college enrollment rate due to the reduced skill premium. The resulting lower share of skilled workers leads to a lower output.

Next, recall that the direct effect can be considered the short-run effect of the introduction, where other macroeconomic variables, such as prices, tax rates, skill distribution, and age distribution, are fixed as in the benchmark. In the short run, aggregate savings and labor supply decrease due to higher fertility; having a child requires parents to spend a fraction of their time and some additional money, reducing their working hours and savings. Thus, the aggregate output decreases in the short run.

Welfare: The direct effect explains more than half of the welfare gains. Importantly, introducing the grants makes agents attend college with smaller costs and enables someone who could not attend college in the benchmark to do that, bringing them a higher lifetime income. The rest of the gains are explained by the distributional effect. The education subsidy increases the share of college graduates in the long run, and the lifetime

utility for college graduates is, in principle, greater than that for high school graduates, particularly because of their higher income due to the college education returns. Thus, this distributional change increases the expected lifetime utility. Note that the direct effect is about the change in $V_{j=1}^P$ in equation (24), while the distributional effect is about the change in μ there.

E.(iii) Expansion

In this section, I consider the effects of raising the income threshold $\bar{I}$ so that students in households of broader income classes can be eligible. Recall that I set the threshold $\bar{I}$ so that the existing grants target the students in households at the bottom 15 % of the income distribution. In this experiment, I increase $\bar{I}$ so that it corresponds to the 40, 50, and 60 percentiles of the income distribution and solve the stationary equilibrium in each case. For students in the household at the bottom 15 %, the payment is still given by g , which amounts to two-thirds of the average expenses of students in the benchmark. For students in households higher than the 15 percentile but less than x percentile of the income distribution, where x takes either 40, 50, or 60, the payment is given by $g/2$, which amounts to one-third of the average expenses of students. In other words, the payment tapers off in income. Letting $\bar{I}_{15\%}$ and $\bar{I}_{x\%}$ denote the income level of 15 and $x \in \{40, 50, 60\}$ percentiles of the income distribution, the grant function $\bar{g}(I; x)$ with a threshold $\bar{I}_{x\%}$ can now be formulated as:

$$\bar{g}(I; x) = \begin{cases} g & \text{if } I < \bar{I}_{15\%} \\ g/2 & \text{if } I \in [\bar{I}_{15\%}, \bar{I}_{x\%}) \\ 0 & \text{otherwise} \end{cases} \quad (25)$$

Table 15 summarizes the results, and there are several takeaways there. First, the equilibrium college enrollment increases as the income threshold is higher. Setting the threshold at the 40, 50, and 60 percentiles of the income distribution, the enrollment rate increases by 4.7 p.p., 5.6 p.p., and 6.2 p.p., respectively, in stationary equilibrium. This is also the case for educational mobility and labor income tax rates. The expansion requires additional revenue, so the equilibrium tax rate should also increase by 0.3 p.p. for the case of the 60% threshold.

However, the expansion would not significantly increase the TFR, as Fig A.9 indicates. Fertility rates for each skill help us understand this situation. First, college graduates continue to increase fertility. This result is straightforward to understand given that the direct and GE effects contribute to the fertility increase of college graduates when the grants are introduced, as discussed in Sections 7.1 and E.(ii). On the contrary, fertility among high school graduates decreases with expansion, making the TFR remain almost constant even though fertility among college graduates increases. Fig A.9 depicts the changes in the TFR and fertility rates for high school and college graduates, where the equilibrium TFR with the existing program (with the 15 percentile threshold) is normalized to one.

	Threshold			
	15%	40%	50%	60%
CL share (Δ p.p.)	+4.0	+4.7	+5.6	+6.2
HS $\rightarrow$ CL (Δ p.p.)	+2.5	+2.6	+3.1	+3.3
Tax (Δ p.p.)	+0.02	+0.17	+0.23	+0.30
Output ($\Delta\%$)	+0.7	+1.3	+0.2	-0.5
Welfare (%)	+5.1	+6.5	+6.3	+5.5

Table 15: Main results of higher income thresholds. *Note:* Rows “CL share” and “HS $\rightarrow$ CL” represent the changes in college enrollment rate and educational mobility in the sense of probability that children of high school graduates attend college. Output changes are expressed as percentage changes. Changes in college enrollment rate, educational mobility, and tax rate are represented as changes in percentage points. Welfare gains are represented in terms of consumption equivalence.

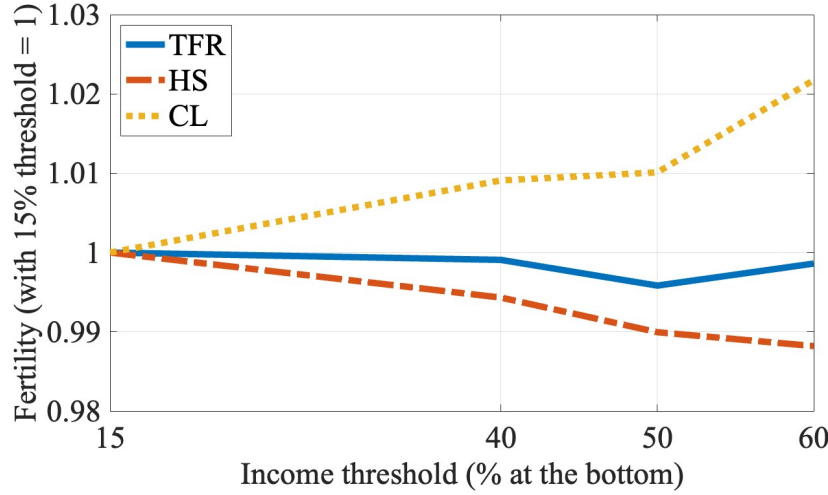


Fig A.9: Changes in fertility rates with expansion. *Note:* The equilibrium TFR with the existing program (with the 15% threshold) is normalized to one. “HS” and “CL” represent fertility rates for high school and college graduates.

Why do high school graduates decrease fertility as the income threshold is higher? To understand this, I conduct the decomposition when the grant function is given by $\bar{g}(I; 60)$, and the results are summarized in Table 16. Two effects are critical to explain the fertility decreases of high school graduates: the direct and GE effects.

In this case with the grant function $\bar{g}(I; 60)$, the wage rate for high school graduates, w_{HS} , is 0.6% higher in the long run than in the benchmark, especially because of the significant supply of college graduates. This higher wage rate makes the opportunity costs of having children for high school graduates higher, which puts downward pressure on their fertility rates.

Next, the direct effects imply a lower fertility rate for high school graduates. This result is somewhat confusing given that we discuss the insurance effects of the grants on fertility, which is critical to account for the fertility increases of college graduates. Why do the direct effects lead to lower fertility for high school graduates? In the benchmark without grants, children of high school graduates are unlikely to attend college. If the grants are introduced and their target expands, educational mobility increases in the sense that

children of high school graduates are more likely to attend college than in the benchmark, as represented in Table 15. Given that the grants are not generous enough to cover 100% of the costs to send their children to college, this higher probability of children going to college implies that those parents are more likely to have to make additional transfers upon children’s college enrollment. The expansion thus increases the expected costs of children for high school graduates, which lowers their fertility.

	Direct	GE	Tax	Dist.	All
HS ($\Delta\%$)	-0.7	-1.4	0.0	0.0	-0.8
CL ($\Delta\%$)	+9.1	+4.3	+2.4	+0.6	+13.2

Table 16: Decomposing the effects on fertility with $g(h, I, 60)$. *Note:* Rows “HS” and “CL” represent the percentage changes in fertility rates for high school and college graduates.

Last, the output increases in expansion up to the case of the 40 percentile threshold, but they start declining after that point, as Table 15 and Fig A.10 indicate. If the income threshold becomes sufficiently high, households with children have less incentive to save to support their college enrollment because they will likely be eligible for the grants. This higher probability of eligibility also causes an income effect on labor supply. Thus, the aggregate capital and labor supply decrease when the income threshold becomes sufficiently high, reducing the output. This concavity of output gains with expansion also applies to the welfare gains. The marginal utility gains of the grants decrease in the values of the income threshold while the tax rate increases with expansion. Then, the net gains start declining at some point.

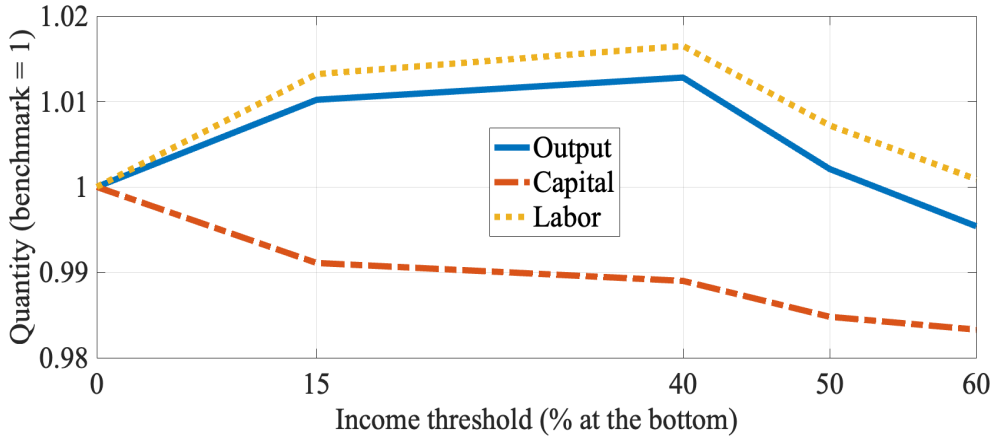


Fig A.10: Changes in aggregate output, capital, and labor supply in efficiency unit with expansion. *Note:*

E.(iv) Comparison With Pro-natal Transfers: Macroeconomic Effects

This section examines the differences in the macroeconomic implications between the education subsidy and child benefit (i.e., typical pro-natal cash transfers).⁵⁵ To this end, I simulate the expansion of the unconditional cash transfers to households with children

⁵⁵In Section E.(v), I also consider reallocating resources into different targets to implement the education subsidy by considering unconditional transfers to college students and introducing ability tests.

B from its benchmark value while setting $\bar{g}(I) = 0$ for any I as in the benchmark. The new level of the per-child payment B is set so that the short-run expenditure upon the introduction is the same as that of the existing income-tested grants, which leads to a 6.5% increase in the per-child payment.

I solve the stationary equilibrium with the expansion of the per-child payment B , and the main results are summarized in Table 17. Columns “Education” and “Typical” represent the results of the introduction of income-tested grants and the expansion of child benefit (“typical” pro-natal transfers).

The child benefit expansion leads to a 2.5% increase of the TFR in the long-run equilibrium, comparable to introducing the income-tested grants but slightly lower than that. The previous decomposition analysis suggests that targeting students in unlucky households in which negative income shocks are realized would be effective, at least marginally, in increasing the TFR. A notable observation is that the characteristics of households who respond to the policy differ between the two cases; the education subsidy induces skilled households to have more children, while the child benefit has a similar degree of impact on skilled and unskilled households regarding fertility. This is because the former is more beneficial for households whose potential children are likely to attend college, and college graduates tend to be those households due to the intergenerational persistence of education.

The college enrollment rate does not change in response to the child benefit expansion. The expansion also does not affect the average human capital, defined as the workers’ average of labor efficiency $\eta_{j,z,e,h}$,⁵⁶ while the education subsidy increases the average human capital by 1.0% primarily because of the higher college enrollment rate. The difference in the average human capital also leads to the difference in the per-capita output. While the education subsidy increases the per-capita output by 0.7% in the long run, the child benefit expansion would rather lead to a 0.3% lower per-capita output. As in the case of introducing the education subsidy, the child benefit expansion also leads to lower (physical) capital accumulation due to crowding out and demographic change. The education subsidy has sufficiently large positive effects on aggregate labor supply in efficiency units, which surpasses the negative impacts on physical capital accumulation and increases output. The child benefit expansion has more modest impacts on aggregate labor efficiency than the education subsidy, leading to the gap in output gains between the two policies.

The child benefit expansion does not affect the standard deviation of wages, whereas the education subsidy leads to a lower standard deviation by enabling some students in poor households to attend college and acquire skills. Lastly, the expansion leads to a 0.3% of the welfare gain, which is substantially lower than a 4.8% gain with the education subsidy. As discussed in Section E.(ii), the welfare gains from the education subsidy primarily stem from higher expected lifetime income across states and an increased probability of attaining a college degree and earning higher wages, driven by higher college enrollment rates and greater educational mobility. In contrast, the expansion of child benefits does not yield similar effects. Thus, the education subsidy leads to greater welfare gains than

⁵⁶Formally, I define the average human capital as $\sum_j \int_{z,h} [(1-s) \cdot \eta_{j,z,e=0,h} + s \cdot \eta_{j,z,e=1,h}] dF(z,h) \mu_j$, where s represents the college enrollment rate, $F(z,h)$ represents the stationary distribution over (z,h) , and μ_j is the stationary distribution of age.

the child benefit expansion under the veil of ignorance.

A caveat is that this model does not capture the endogenous human capital accumulation before high school graduation and the dynamic complementarity of human capital. Households may increase their investments in their children upon the child benefit expansion, which contributes to greater human capital; however, that channel is not considered in this current analysis. Those ingredients can be critical in comparing the effects of these different programs, especially on college enrollment rates, aggregate human capital, and output. The exercise in this framework is a first step, and incorporating those ingredients—fertility, college enrollment, dynamic complementarity of human capital—in one framework is left for future research.

	Education	Typical
TFR ($\Delta\%$)	+3.1	+2.5
HS	−0.3	+2.5
CL	+10.5	+2.4
CL share (Δ p.p.)	+4.0	0.0
HS→CL	+2.5	0.0
Avg. HC ($\Delta\%$)	+1.0	0.0
Output ($\Delta\%$)	+0.7	−0.3
STD (wage) ($\Delta\%$)	−1.3	0.0
Welfare (%)	+5.1	+0.3

Table 17: Main results with several schemes with different targets. *Note:* Columns “Education” and “Typical” represent the results with the introduction of the education subsidy and expansion of child benefit (“typical” pro-natal transfers), respectively. Values in each cell indicate changes from the benchmark value. Rows “HS” and “CL” indicate the percentage changes in the fertility of high school and college graduates. “HS→CL” represent the changes in college enrollment rate and educational mobility in the sense of probability that children of high school graduates attend college. “Ave. HC” stands for the average human capital.

E.(v) Reallocating Resources to Different Programs

This section examines the performance of the potential alternative programs to the existing income-tested grants. To this end, I consider another two scenarios, in addition to the introduction of income-tested grants to the benchmark model. The first scenario is to introduce grants for college students with income and ability tests so that “high-ability” students in poor households are eligible. More specifically, I keep the income threshold adopted in the existing scheme but arbitrarily set the lower bound for students’ innate ability for the eligibility, $\underline{h}$, to its median. The payment function $g(h, I)$ for this scheme is defined as follows:

$$g(h, I) = \begin{cases} g_1 & \text{if } I < \bar{I} \text{ \& } h > \underline{h} \\ 0 & \text{otherwise} \end{cases}$$

Here, the payment g_1 is set so that the short-run expenditure upon the introduction is the same as that of the existing income-tested grants. As a result, g_1 covers approximately 100% of the average student’s expenditure in the benchmark, which is greater than the payment in the existing scheme with only income tests because the number of eligible students is fewer.

The second scenario introduces unconditional grants for college students regardless

of ability and income. The payment function is given as $g(h, I) = g_2$ for any students with (h, I) , where g_2 is set so that the short-run government expenditure is the same as the existing program. As a result, g_2 covers approximately 10% of the average students' expenditure in the benchmark, which is less significant than in the existing income-tested grants because this alternative program covers a broader range of students.

I solve the stationary equilibrium with each scenario, and the results regarding fertility and enrollment rates are summarized in Table 18. A highlight is that the existing scheme with income tests would lead to the highest equilibrium TFR among other scenarios. Notably, the unconditional grants lead to a 1.1 p.p. higher college enrollment rate than the income-tested ones and a -0.3% lower TFR because of the composition effect.

	Income	+Ability	Uncond.
CL share (Δ p.p.)	+4.0	+2.6	+5.1
TFR ($\Delta\%$)	+3.1	+2.7	-0.3
HS	-0.3	-3.5	+4.0
CL	+10.5	+8.4	+3.9
Output ($\Delta\%$)	+0.7	-0.1	-0.7
STD (wage) ($\Delta\%$)	-1.3	-0.9	-0.1
Welfare (%)	+5.1	+2.6	+1.6

Table 18: Main results with several schemes with different targets. *Note:* values in each cell indicate changes from the benchmark value. Rows “HS” and “CL” indicate the percentage changes in the fertility of high school and college graduates.

E.(vi) Other Policies Insuring against the Risks

This section studies two other policies that potentially insure against the risks associated with fertility choices under income uncertainty, other than the income-tested subsidy examined in Section 7: (1) wage insurance that provides cash benefits with households with a negative income shock and (2) fertility-dependent wage insurance that provides wage insurance benefits with households depending on the number of children they have.

Wage insurance provides workers facing negative income shocks (i.e., $z < 0$) with cash transfers. In the uniform wage insurance, I introduce the following payment function into the budget constraint for workers:

$$w_I(z) = \begin{cases} w_I & \text{if } z < 0 \\ 0 & \text{otherwise} \end{cases}$$

For the fertility-dependent wage insurance, I introduce the following payment function into the budget constraint for workers:

$$\tilde{w}_I(z, n, j) = \begin{cases} n \cdot \tilde{w}_I & \text{if } z < 0 \text{ \& } j < J_{IVT} \\ 0 & \text{otherwise} \end{cases}$$

Under the fertility-dependent wage insurance, workers facing a negative income shock receive wage insurance benefits proportional to the number of children younger than 18 that they have.

I implement the introduction of these policies in an expenditure neutral way. More specifically, I feed the income-tested education subsidy into the model as I do in Section

7, and then set policy parameters for each cash transfer so that the short-run government expenditures of each policy is the same as those of the income-tested education subsidy. Also, I simulate each policy in a partial equilibrium setup to focus on the direct effects on fertility, controlling for indirect effects such as a GE effect and distributional effect.

The results are summarised in Table 19. Introducing fertility-dependent wage insurance increases fertility by 0.9%, whereas introducing uniform wage insurance has no effect on fertility. The insurance effects of the income-tested college subsidy on fertility, which result in a 1.3% increase in fertility, are more significant than those of these policies. This is because the income-tested college subsidy provides partial insurance not only against income risk but also against another source of uncertainty: uncertainty regarding children’s characteristics (i.e., psychic costs and innate abilities) and the associated uncertainty regarding the willingness to pay for children’s education.

	Fertility ($\Delta\%$)
Income-test college subsidy	+2.1%
Wage insurance (uniform)	+0.0%
Wage insurance (fertility)	+0.9%

Table 19: Changes in fertility with policies that insure against the risks.

E.(vii) Transitional Dynamics: Additional Figures

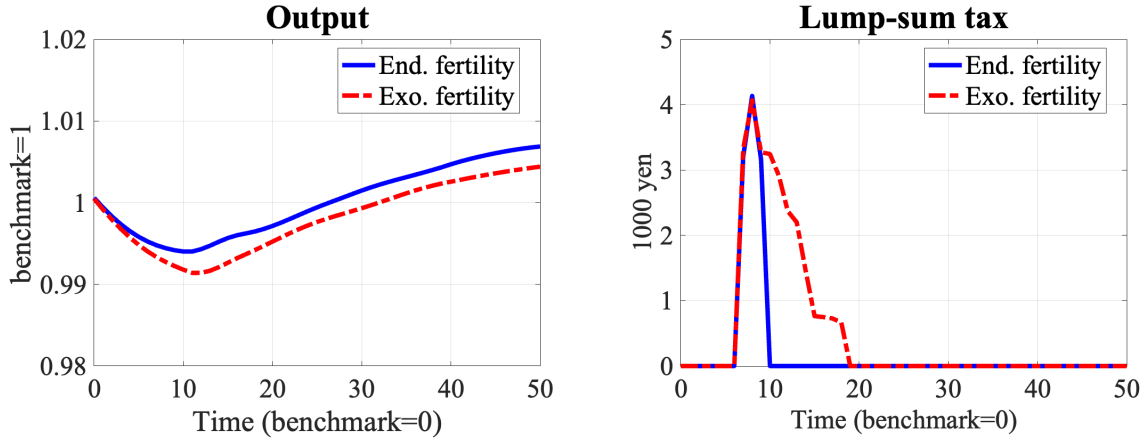


Fig A.11: Transitional dynamics of output and lump-sum tax. *Note:* “End. fertility” and “Exo. fertility” represent transition paths under endogenous fertility and exogenous fertility.

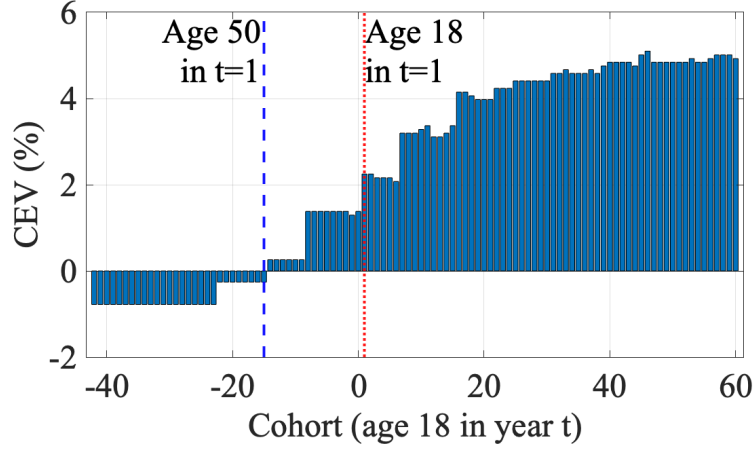


Fig A.12: Welfare effects for each cohort along the transition. *Note:* The horizontal axis represents the cohort, where larger values mean younger cohorts.

F Empirical Analysis

In Section 7, I show that income-tested grants provide partial insurance against income risk and the associated volatility in utility when households have a (or an additional) child. This section presents suggestive empirical evidence supporting this result, exploiting variation across contract types in line with a strand of the empirical literature.

More specifically, I follow the recent work of [Guner et al. \(2024\)](#), who examine differences between permanent and temporary contracts in Spain and show that workers under temporary contracts—characterized by higher probabilities of job destruction—have lower birth probabilities. Applying their strategy is appropriate in this context because a similar institutional framework exists in Japan, where the distinction is between so-called regular and irregular workers, with the latter facing a higher probability of job loss, as described below.

To isolate the effect of income uncertainty on fertility as cleanly as possible, I extend the reduced-form analysis of [Guner et al. \(2024\)](#) in two main ways. First, I incorporate the individual fixed effect to account for time-invariant unobserved heterogeneity in fertility decisions, thereby partially addressing selection concerns related to irregular contracts. Second, I exploit a survey question from the JPSC that asks respondents about the number of additional children they would like to have, which allows me to measure fertility intentions directly. By exploiting within-individual variation in this measure, I attempt to account for an important component of unobserved time-varying heterogeneity—namely, fertility preferences.

I find that, after controlling for household characteristics—including household income, household assets, the number of existing children, fertility intentions, and age—and exploiting within-individual variation in contract type, irregular employment is associated with a 1.3 percentage point lower probability of giving birth in the following year relative to regular employment. However, because this finding does not directly establish a link between income uncertainty and fertility, I attempt to account for alternative mechanisms in the regression and discuss the interpretation of the results accordingly. Ultimately, the evidence suggests that income uncertainty may constitute an important mechanism underlying the negative association between irregular employment and fertility.

I begin by reviewing the empirical literature on the effect of income uncertainty on fertility. I then provide institutional background and summary statistics by contract type in my sample based on the JPSC. Finally, I estimate regressions examining the association between temporary contracts and birth probabilities and discuss how the results can be interpreted as reflecting the effect of income uncertainty.

Literature: Providing direct evidence on the relationship between income risk and fertility is often challenging, primarily because it is difficult to construct a direct measure of perceived income uncertainty and to obtain plausibly exogenous variation in such a measure. Nevertheless, previous empirical studies examine the link between income uncertainty and fertility in at least two ways.

The first strand of the literature relies on aggregate measures of economic uncertainty and documents a negative association between uncertainty and fertility, typically exploiting cross-country or cross-state variation in measures of uncertainty and fertility rates. For example, [Gozgor et al. \(2021\)](#) use cross-country panel data with the World Uncertainty Index (WUI), compiled by the International Monetary Fund, and find that higher uncertainty is associated with lower fertility. It is important to note, however, that the WUI reflects several dimensions of uncertainty, such as those related to domestic policy and international trade, and does not directly capture income uncertainty. Relatedly, [Comolli and Vignoli \(2021\)](#) focus on Italy’s sovereign debt crisis of 2011–2012, construct a measure of uncertainty using Google Trends data for the term “spread,” and show—based on a regression discontinuity design—that an increase in perceived uncertainty is associated with lower birth rates. [Chabé-Ferret and Gobbi \(2018\)](#) exploit variation across U.S. states and over the course of the twentieth century in economic uncertainty, measured by the volatility of income growth across states and years. They show that economic uncertainty reduces cohort fertility and that the decline in uncertainty accounts for a significant fraction of the post–World War II baby boom.

The second strand of the literature exploits variation in contract types, which is informative about differences in layoff probabilities, as well as policy reforms that affect workers with different contract types (e.g., [Prifti and Vuri, 2013](#); [De Paola et al., 2021](#)). For example, [De Paola et al. \(2021\)](#) exploit a labor market reform in Italy that reduced job security for workers with open-ended contracts and show that the reform reduced fertility among the treated workers. It is important to note, however, that such reforms are likely to affect not only income volatility (a second-moment effect) but also expected income levels (a first-moment effect). Another set of studies along these lines uses variation in contract types—informative about job security—to examine their association with fertility ([Modena et al., 2014](#); [Guner et al., 2024](#)). For example, [Guner et al. \(2024\)](#) examine differences between permanent and temporary contracts in Spain and show that temporary workers—who face higher layoff probabilities—are less likely to have a child.

Below, I follow the second strand of the literature to provide supporting evidence for the model-based results, in particular building on [Guner et al. \(2024\)](#). The institutional setting is similar to that of Spain studied in [Guner et al. \(2024\)](#). Although variation across contract types in principle captures factors other than exogenous variation in income uncertainty, I carefully specify the empirical framework to isolate the effects of income uncertainty as much as possible, as discussed below.

Institutional background and descriptive statistics: The Japanese labor market is characterized by a dual structure consisting of two main contract types: regular and irregular employment. This dual structure originated in Japan’s postwar employment system, under which regular workers were granted strong job protection and seniority-based wages. Following the economic slowdown in the 1990s, firms increasingly relied on irregular workers as a flexible margin of adjustment, a trend facilitated by a series of labor market reforms that expanded the permissible use of fixed-term and dispatched workers. As a result, differences in job security between regular and irregular workers became institutionalized and continue to play an important role in the Japanese labor market.

These two types differ along several dimensions, including hourly wages, working hours, and layoff probabilities. Regular contracts offer higher wages and stronger job security but require longer working hours with less flexible schedules. In contrast, irregular contracts offer greater flexibility with fewer working-hour commitments, but are associated with lower pay and weaker employment protection. Women are more likely to work under irregular contract, while men (particularly married men) are mostly work under regular contracts.

To see these differences, I focus on married women born in 1959–69 in JPSC. Table 20 summarizes descriptive statistics on irregular employment, including the share of irregular workers by age and education, separation rates, hourly wages, working hours, and birth probabilities.

On average, 24.6% of married women in these cohorts are employed under irregular contracts. The share is lower—about 15%—among those under 30 and rises with age. In this cohort, many married women leave the labor force around childbirth and later reenter employment in irregular jobs, which offer more flexible and less time-demanding work schedules. Irregular employment is common among women but not among men: on average, only 2.24% of married men work under irregular contracts.

As discussed above, irregular contracts are characterized by (1) higher separation rates, (2) fewer working hours, and (3) lower hourly wages. First, about 3.87% of irregular workers experience involuntary job loss each year, compared with 0.86% for regular workers. Second, among women aged 30–39, average weekly hours under irregular contracts are 26.68—about 30% lower than for regular workers (38.13). Third, hourly wages for women aged 30–39 under irregular contracts are likewise about 30% lower than for their regular-contract counterparts.

Finally, the annual birth probability for irregular workers is 0.0091, substantially lower than the 0.0792 observed for regular workers ($\simeq 0.91\%$ vs. 7.92%). This pattern holds for both college and non-college graduates to a similar extent. These average birth probabilities indicate that irregular workers are less likely to give birth in a given year.

	Average	Non-college	College
<i>Share of irregular workers</i> (conditional on working and on education)			
Women			
Age 30 >	0.1515	0.1540	0.1176
Age 30–39	0.2637	0.2795	0.1327
Age 40–49	0.2724	0.2817	0.2011
Average	0.2464	0.2598	0.1291
Men (Average, Age < 50)	0.0224	0.0306	0.0070
<i>Job separation rate (annual)</i>			
Regular (women)	0.0086	0.0092	0.0
Irregular (women)	0.0387	0.0353	0.0909
<i>Working hours per week</i>			
Regular (women, age 30–39)	38.13	37.67	40.61
Irregular (women, age 30–39)	26.68	26.93	22.95
<i>Hourly wage (Irregular & non-college = 1)</i>			
Regular (women, age 30–39)	1.495	1.438	1.805
Irregular (women, age 30–39)	1.025	1.000	1.394
<i>Birth probability</i>			
Regular (women)	0.0792	0.0786	0.0801
Irregular (women)	0.0091	0.0118	0.0026

Table 20: Descriptive statistics on regular and irregular workers in Japan. *Note:* The sample is restricted to married women born in 1959–1969. Ages are limited to those younger than age 50, which I take as the upper bound of reproductive age. Hourly wages for non-college graduates (College >) employed under irregular contracts are normalized to 1.

Specification: To examine the relationship between irregular contracts and fertility while controlling for potential confounding factors, and to investigate the underlying mechanism, I estimate the following linear probability model using the JPSC sample:⁵⁷

$$y_{i,t} = \alpha + \beta T_{i,t-1} + \mathbf{x}_{i,t-1}\boldsymbol{\theta} + \psi_i + \phi_t + \varepsilon_{i,t}. \quad (26)$$

Here, $y_{i,t}$ takes one if the wife in a couple i gives birth in year t . $T_{i,t-1}$ is a dummy variable taking one if the wife worked under an irregular contract in year $t - 1$, and β is the coefficient of our interest. Using the lagged indicator helps mitigate concerns about reverse causality—namely, that a woman may sort into (or out of) irregular employment in response to having given birth. The vector $\mathbf{x}_{i,t-1}$ includes other couples’ characteristics observed in $t - 1$, such as household income, household assets, the number of existing

⁵⁷The sign and statistical significance of the coefficient of interest remain unchanged when I estimate the model using a logistic specification, as in [Guner et al. \(2024\)](#). I nevertheless adopt the linear probability model as my benchmark specification because its coefficients are more straightforward to interpret.

children, wife’s age, wife’s labor force participation, husband’s contract type, and *fertility intention*—the number of children she would like to have additionally. ψ_i and ϕ_t denote individual (couple) and year fixed effects.

Results: Table 21 reports the regression results. Column (1) presents estimates without fixed effects. Because the specification also includes an indicator equal to one if the woman did not participate in the labor market in the previous year, the coefficient on the irregular contract indicator captures the effect of irregular employment relative to regular employment. The estimate suggests that, holding other variables—such as income, assets, age, number of children, fertility intention, and education—constant, being an irregular worker is associated with a 2.3 percentage point lower probability of giving birth relative to regular workers. The coefficient is statistically significant at the 1 percent level.

Columns (2) and (3) gradually add couple and year fixed effects. The coefficient remains negative, although its statistical significance declines; it remains marginally significant after including both sets of fixed effects. More specifically, column (3) implies that within-individual transitions into irregular employment are associated with a 1.3 percentage point lower probability of birth, conditional on the number of existing children, assets, and fertility intentions. The decline in magnitude and statistical significance suggests that selection based on unobserved individual characteristics partly explains the negative association between irregular employment and fertility. Nevertheless, the coefficient remains marginally significant, suggesting that selection based on unobserved heterogeneity alone cannot fully account for the observed relationship. This raises the question of which differences between regular and irregular contract types drive the observed fertility differential.

	(1)	(2)	(3)
Irregular _{<i>t</i>−1}	−0.023*** (0.006)	−0.015** (0.007)	−0.013* (0.007)
Individual FE	No	Yes	Yes
Year FE	No	No	Yes
Observations	5042	5042	5042

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 21: Regression results for birth probabilities among married women aged 24–49

Discussion and interpretation of the results: As discussed earlier, irregular and regular contracts differ along multiple dimensions, making it unclear which specific component drives the observed relationship. The main differences concern (1) working hours, (2) hourly wages, and (3) job security. Along the working-hours margin, standard models of fertility with time costs would predict that irregular contracts could increase fertility by relaxing time constraints. This prediction is qualitatively inconsistent with the regression results, suggesting that time flexibility is unlikely to be the primary mechanism behind the negative association.

By contrast, the lower wages and earnings typically associated with irregular em-

ployment may reduce fertility through a negative income effect, potentially mimicking an “income uncertainty” channel rather than reflecting income volatility per se. To address this concern, I include household income and asset levels as controls. Although these variables do not perfectly capture expected lifetime income—which is arguably the more relevant concept—they partially account for differences in economic resources across contract types. The regression results show that neither income nor asset levels have statistically significant explanatory power.

Taken together, my regression analysis demonstrates that within-individual variation in contract type is negatively associated with the probability of birth, after accounting for time-invariant household characteristics, time-varying fertility intentions, and other controls such as income, assets, and age. These findings suggest that factors not directly captured in the specification—particularly differences in income uncertainty between regular and irregular contracts—may constitute an important mechanism underlying the negative association between irregular employment and fertility.